Highlights

Tonal-Perceptual Caching for On-Device Vietnamese Speech Translation

Anonymous Author

- Two-stream AND-existence cache: 32-d learned tone embedding (26 KB) plus 260-d segmental stream.
- $92/100$ tonal correct-miss at $\tau_A = 0.85$, $100/100$ at $\tau_A = 0.90$ under 10 dB noise, on $N = 120$.
- Cache-hit latency 25 ms, 1,441× over FP32 cold cascade on short-utterance representative runs.
- Parallel diacritic-preserving T2 hash cuts text-cache collisions 95.8× (16.48% → 0.17%).
- Syllable-local tone pooling cuts adversarial sentence-level false-hit from 46.7% to 16.7% training-free, 3.3% with a trained pooling head.

Tonal-Perceptual Caching for On-Device Vietnamese Speech Translation⋆

Anonymous Author[a,*,1]

[a]*Anonymous Institution, , , , ,*

ARTICLE INFO

Keywords:
on-device speech translation
low-resource languages
tonal languages
Vietnamese
audio fingerprinting
edge AI
quantization
utterance caching

ABSTRACT

On a 120-clip Vietnamese tonal minimal-pair corpus (20 carrier syllables × 6 tones, MMS-TTS-vie synthesis), a contrastively learned 32-d tone embedding (26.05 KB on disk; same-syllable-different-tone hard negatives) coupled with a 260-d segmental stream and gated by an AND-existence voting rule reaches 92/100 **correct-miss on tonal variants at** $(\tau_A, \tau_B) = (0.85, 0.93)$ and 95/100 at $\tau_A = 0.90$, both at 120/120 self-lookup recall, saturating at 100/100 at $\tau_A = 0.90$ under DEMAND-like additive noise at SNR $\in \{10, 5\}$ dB. This is a +56 pp absolute lift over the same gate driven by a heuristic 6-bin tone histogram, exposing the prior 36% ceiling as a tonal-representation limit rather than a voting-rule limit. On cache-hit, end-to-end latency drops to 25 ms, **1**,**441**× over the 36.5 s FP32 cold cascade on representative single-utterance runs (the ratio is short-utterance peak; for longer inputs the cache-hit pathway preserves order-of-magnitude wins). The cache short-circuits ASR, MT, and TTS entirely on hit. A parallel diacritic-preserving hash on the T2 text cache, AND-gated with a diacritic-stripped stream, reduces semantic collisions **95**.**8**× (from 16.48% to **0**.**17**%) on a 3,000-transcript audit at zero paraphrase-recall cost (20/20). Sentence-level adversarial Vietnamese minimal pairs, which previously presented a 46.7% false-hit failure mode under utterance-level tone pooling, drop to **16**.**7**% under a training-free syllable-local max-pool over the same 32-d learned tone embedding (and to **3**.**3**% when the pooling head is itself contrastively trained on the adversarial pairs), confirming that the residual was a pooling-locality limit rather than a representation limit. The full INT8 cascade (PhoWhisper-small + NLLB-200-distilled-600M + multilingual VITS) fits in 861 MB and runs CPU-only. Tone-bin structure is Vietnamese-orthography-aligned; cross-language reuse requires per-target re-binning (Mandarin: 5, Thai: 5, Yoruba: 3).

1. Introduction

A translation cache that confuses *ma* (ghost) with *má* (mother) is not one percent wrong; it is wrong in a way that no industrial user can trust. The contrast extends through the full Vietnamese tone inventory: *mạ* means tomb, *mã* means horse, and *mà* is a contrastive conjunction. Every one of these distinctions rides on a single suprasegmental contour that segmental acoustic features compress away. Two acoustic realizations that differ only in tone are, to a music-style perceptual hash (Wang, 2003), indistinguishable; to a downstream worker reading the translated output on a wrist-mounted display, they are six unrelated sentences. This is the failure mode that motivates the present work.

The deployment scenario sharpens the problem. A recent industrial speech-translation challenge targets factory floors, construction sites, and hospital wards, where workers' hands are occupied, network connectivity is intermittent or absent, and ambient noise is loud and persistent. The defining property of speech in such environments is repetition: most of what a worker says during a shift is a short directive drawn from a small canonical vocabulary, safety calls, equipment names, location markers, uttered again and again by the same speaker and re-uttered across the rest of the crew.[1] That regularity is the lever. If the same phrase has been translated once, it should not be translated again, and the cost of recognizing it the second time should be orders of magnitude lower than the first.

The existing cache primitives, however, all fail in this setting. Exact byte-level hashing of raw audio achieves a hit rate of zero on real speech, because no two utterances, not even repeats by the same speaker, are bit-identical after analogue capture, voice activity detection, and chunking. Generic perceptual audio hashes, descended from the music identification literature (Wang, 2003), achieve high robustness to additive noise but are explicitly engineered to discard the linguistic content that defines tonal contrast; on the six-way Vietnamese minimal-pair set, off-the-shelf perceptual hashes collapse all six tones into a single bucket. Neither family was designed for the cache primitive on-device translation actually requires: a fingerprint whose collision structure tracks phrase-level *semantic* identity in a tonal language. The recognition layer is well understood, with PhoWhisper (Le, Nguyen and Nguyen, 2024) providing a strong Vietnamese ASR baseline, but the caching layer above it is unaddressed.

⋆Submitted to Information Processing & Management, Special Issue "Multilingual and Multimodal Low-Resource Language Processing" (VSI: MMLowRes). Double-blind submission.

*Corresponding author
anonymous@example.org (A. Author)
ORCID(s):

[1]Author identifying details withheld for double-blind review.

[1]Field surveys we draw on report 40–90% of utterances per shift as canonical directives. The target hardware is a Qualcomm Snapdragon 8 Gen 3 with a Hexagon NPU; the application envelope is a sub-1.5 s end-to-end latency budget within 4 GB of NPU memory, with no cloud round-trip permitted.

If those claims hold, an entire class of industrial deployments, currently blocked on either cloud latency or on-device compute, becomes feasible on hardware that already fits in a worker's pocket.

Contribution roadmap. We close the gap with two clusters of contributions. The *first cluster* is what is novel about the fingerprint itself. (i) A 272-dimensional tonal-perceptual fingerprint built from five linguistically grounded blocks (a 6-d Vietnamese tone histogram, a 130-d temporal-subsampled MFCC trajectory, a 130-d first-difference delta, 4-d F0 statistics, and 2-d energy statistics) engineered so that two recordings of the same phrase hash together while two tonal minimal pairs hash apart. (ii) A two-stream voting cache that treats the 6-d tone histogram and the 260-d segmental remainder as independent fingerprints and requires *some* stored entry to pass both stream-specific thresholds before issuing a hit; on a Vietnamese tonal-minimal-pair corpus this achieves a 72% correct-miss rate at the present $\tau_A = 0.85$ operating point, where every single-vector concatenation we tried collapses to uniform false hits.

The *second cluster* is what makes the design deployable. (iii) An INT8 cascade substrate (faster-whisper, CTranslate2-NLLB, dynamically quantized VITS) that compresses the FP32 cascade by 5.4× on CPU and, when the cache hits, by a further three orders of magnitude (25 ms end-to-end on a representative single-utterance run, 1,441× over the cold FP32 cascade). (iv) A typology-aware voice activity detector that gates on F0 stability rather than energy, recalibrated to the natural Vietnamese pitch range after a generic-speech default rejected the majority of in-domain audio. (v) A reproducible log of four abandoned alternatives (an MFCC speaker cache, coarse-to-fine MT routing, weighted-concatenation fingerprints, PCA-whitened single-vector fingerprints) that collectively rule out single-vector fixes and identify two-stream voting as the load-bearing design decision.

The remainder of the paper is organized as follows. Section 2 surveys the on-device speech-translation, mixed-precision quantization, audio-fingerprinting, and tonal-speech literatures and locates the missing primitive precisely. Section 3 formalizes the fingerprint and the two-stream voting architecture. Section 4 reports the fingerprint-component ablation, the tonal-minimal-pair evaluation, end-to-end latency, precision-recall calibration, cache scalability through $N = 1024$ entries, and cross-language generalization to Mandarin, all with bootstrap confidence intervals over $\geq$ 1000 resamples. Section 5 reports the on-device memory and latency profile. Section 6 expands the negative results. Section 7 concludes.

2. Related Work

2.1. On-device speech translation

Speech-to-speech translation has converged on two architectural families. Cascaded systems compose an automatic speech recognizer, a neural machine translator, and a text-to-speech synthesizer, typically instantiated with multilingual backbones such as Whisper for recognition and NLLB-200 for translation (NLLB Team, Costa-jussà et al., 2022). End-to-end systems collapse the cascade into a single model, with SeamlessM4T (Seamless Communication, Barrault et al., 2023) the current reference point for high-resource directions.

Both families were originally designed for cloud inference, and their on-device viability rests on a downstream engineering substrate. The CTranslate2 runtime (Klein, Hernandez, Nguyen and Senellart, 2020) and its faster-whisper (SYSTRAN, 2023) wrapper provide INT8-quantized Transformer inference with custom CPU kernels, while GGML-style quantization (Gerganov and GGML contributors, 2023) ports the same idea to mobile-class CPUs through aggressively packed 4-bit and 5-bit tensor layouts. These engineering tools, however, are language-agnostic: their kernels treat every layer and every language uniformly.

As a consequence, the on-device speech-translation literature has concentrated on high-resource pairs where degradation under uniform quantization is acceptable, and reports for tonal low-resource languages such as Vietnamese remain sparse. The combination that this work addresses, namely end-to-end Vietnamese translation, fully on-device, with tone-preserving quality on industrial speech, is largely absent from existing benchmarks, and no public system reports a deployment-grade latency profile under both noise and tonal-minimal-pair stress.

2.2. Mixed-precision quantization for speech models

The general quantization literature provides three load-bearing techniques for compressing Transformer-based speech and language models: GPTQ (Frantar, Ashkboos, Hoefler and Alistarh, 2023) approximates a layer-wise reconstruction objective with second-order information, SmoothQuant (Xiao, Lin, Seznec, Wu, Demouth and Han, 2023) rebalances activation outliers into weights so that both can be quantized to INT8, and AWQ (Lin, Tang, Tang, Yang, Chen, Wang, Xiao, Dang, Gan and Han, 2024) selects salient weight channels by activation magnitude and protects them at higher precision. All three are linguistically blind: their saliency criteria depend exclusively on activation statistics observed on a calibration set.

The recent speech-specific line tightens this further. Xu, Li, Jin, Wang, Chen, Li, Geng, Hu, Deng and Liu (2025) apply a Gumbel-Softmax differentiable architecture search over per-layer bitwidths from $\{2, 4, 8\}$ and report the strongest Pareto frontier currently available for ASR compression. The architecturally most similar competitor, FADE (Wang, Zhao, Luo, Wu, Ma, Tian, Ding, Chang and Lu, 2026), allocates precision by a diagnostic coefficient derived from frequency-domain perturbation sensitivity. TAQ (LeVi, Lapid, Himelstein, Baskin, Shwartz Ziv and Mendelson, 2025) introduces task-aware quantization

differentiation for Mandarin, the closest published acknowledgment that tone matters for compression.

None of these prior systems select precision by the *linguistic property* the layer encodes; their criteria are statistical (AWQ, SmoothQuant), gradient-based (Xu et al.), or diagnostic-loss-based (FADE). Typology-aware allocation, which uses a tone-probe to identify which layers carry the linguistic information that quantization must protect, remains an open design point.

2.3. Audio fingerprinting and perceptual caches

The audio fingerprinting tradition originates with Shazam-style spectral peak hashing (Wang, 2003), which compresses an audio clip into a sparse landmark set engineered for robustness to additive noise and partial overlap. Neural extensions, beginning with the Million Song Dataset embeddings (Bertin-Mahieux, Ellis, Whitman and Lamere, 2011) and continuing through contrastive audio encoders, replaced hand-engineered landmarks with learned representations but preserved the same target task: identifying which *recording* a query came from, with no requirement to preserve linguistic semantics. In a parallel literature, key-value cache compression for large language models (Liu, Yuan, Jin, Zhong, Xu, Braverman, Chen and Hu, 2024) reuses attention state across tokens but operates on internal model representations rather than on audio input. Both lines optimize for a goal orthogonal to translation reuse: music fingerprints discard the lexical content they happen to capture, and KV-cache methods never see the acoustic signal at all. The result is a missing primitive: a fingerprint engineered so that two acoustic realizations of the same *phrase* hash together while two tonal minimal pairs hash apart. To the best of our knowledge, no prior work designs an audio fingerprint whose collision structure tracks phrase-level semantic identity in a tonal language, which is precisely the cache key that on-device translation reuse requires.

2.4. Tonal-language speech processing

Vietnamese-specific speech recognition has consolidated around PhoWhisper (Le et al., 2024), a VinAI fine-tune of Whisper that establishes the strongest publicly reproducible Vietnamese ASR baseline and serves as the backbone of the present system. Complementary recent contributions include VietASR (Zhuo, Yang, Shao, Xu, Yu, Yu and Chen, 2025), which reports competitive WER on conversational Vietnamese, and SeaLLMs-Audio (Liu, Aljunied, Chen, Chan, Xu, Rong and Zhang, 2025), a Southeast-Asian speech-language foundation model that extends multilingual coverage to under-resourced regional languages. The probing literature provides the linguistic anchor: de la Fuente and Jurafsky (2024) localize Mandarin tone information to layer 8 of self-supervised speech encoders, demonstrating that tonal contrasts are not diffusely distributed across the network but concentrate in identifiable mid-encoder regions. This finding directly motivates a tone-histogram component in any fingerprint that aspires to discriminate tonal minimal pairs, and provides empirical justification for treating a small number of acoustic dimensions as carrying disproportionate semantic weight. Prior tonal-ASR work, however, stops at the recognition transcript: it neither exposes a fingerprint suitable for cache-key construction nor exploits the layer-localization result for downstream system design. The contribution of typology-aware *cache key design*, which treats tone inventory as a hyperparameter of the fingerprint dimension, is, to the best of available evidence, novel.

2.5. Industrial speech systems and deployment

A small but growing literature addresses deployment-side concerns for low-resource speech. The closest published scenario is a privacy-preserving Vi-En translation system for iOS (Le, 2025), which establishes both the feasibility of and the user demand for fully on-device Vietnamese pipelines, but reports neither a quantization strategy beyond off-the-shelf CoreML conversion nor a caching layer. TAQ (LeVi et al., 2025) represents the orthogonal optimization axis: it differentiates precision by task sensitivity, demonstrating that uniform quantization is suboptimal even within a single model. These two strands point to complementary rather than competing solutions. Cache-level optimization reuses computation across repeated utterances and is bounded by the repetition rate of the deployment scenario; quantization-level optimization reduces the cost of every uncached utterance and is bounded by the precision floor at which tonal contrasts collapse. Neither subsumes the other, and the present work contributes the cache layer of this composition.

The literature thus leaves two gaps unaddressed: no existing audio fingerprint is engineered for tonal-phrase semantic identity, and no published on-device Vietnamese speech-translation system reports a cache-hit pathway that survives tonal-minimal-pair stress. The method introduced in Section 3 closes both.

2.6. Head-to-head positioning

Table 1 positions the present work against the closest published systems. The comparison is qualitative on the two middle axes (Vietnamese support, tone awareness) and quantitative on the right axis (reported headline number from the cited paper). The key reading is that the present contribution does not compete with any single row of the table; it composes with them. Quantization methods optimize the cost of every uncached utterance; the present cache layer optimizes the cost of every repeated utterance. Neither subsumes the other, but until now nothing in the literature delivered the tonal-language cache primitive that on-device industrial speech translation actually requires.

Table 1
Positioning of the present contribution against the closest prior systems. ✓ = explicit, × = no, *impl.* = implicit. "—" = no headline number stated by the cited paper.

System	Vi	Tone	Reported headline
AWQ (Lin et al., 2024)	×	×	4-bit weight quant; MLSys '24 best paper
GPTQ (Frantar et al., 2023)	×	×	3-/4-bit, OPT-175B in ≈4 GPU-hrs
SmoothQuant (Xiao et al., 2023)	×	×	INT8 W+A, up to 1.56× speedup
FADE (Wang et al., 2026)	×	×	3-/4-bit on Whisper/Moonshine (—)
TAQ (LeVi et al., 2025)	×	impl.	Per-task accuracy–memory gains (—)
Xu et al. (Xu et al., 2025)	×	×	3.5-bit HuBERT, 8.6× compression
PhoWhisper (Le et al., 2024)	✓	impl.	4.67% WER on VIVOS (large)
Vi-iOS (Le, 2025)	✓	×	Functional demo, no tone eval (—)
This work	✓	**expl.**	1,441× cache-hit, 92% tonal correct-miss ($N = 120$)

3. Method

3.1. System overview

Headline. Two cache layers (audio T1, text T2) interpose between three INT8 stages so that a cache hit collapses the cascade to a single fingerprint lookup.

Imagine a worker shouting *dùng dây chuyền* ("stop the production line") for the fifth time today. The first time, the full cascade ran. The fifth time, the cache returns the translated audio in twenty-five milliseconds. The pipeline that makes both modes coexist is a fully on-device cascade with two cache layers interposed between three INT8 model stages.

An incoming microphone buffer first passes through a tonal-language voice activity detector (Section 3.8), which rejects silence, hum, and stationary machinery noise before any model executes. The accepted audio is queried against the tonal-perceptual audio cache (T1, Section 3.3); on a T1 hit the cached target-language audio is returned directly, short-circuiting ASR, MT, and TTS in a single fingerprint lookup. On a T1 miss the audio is transcribed by an INT8 Vietnamese ASR (Section 3.9), and the resulting transcript is queried against the text-level translation memory (T2, Section 3.4); on a T2 hit only the TTS stage runs. On a T2 miss the transcript is translated by an INT8 NLLB-200 (NLLB Team et al., 2022) and synthesized by an INT8 VITS vocoder. Both caches are populated on every miss, so the fifth occurrence of a phrase is the one that benefits. Figure 1 diagrams the data flow.

3.2. Theoretical framework

Two pressures pull in opposite directions on the cache key. It must be *forgiving* enough to recognize the same phrase across speakers, microphones, and background noise, yet *strict* enough to keep tonal minimal pairs (the same syllable carried by different contours) on opposite sides of the threshold. The formalism below makes this tension precise so that Sections 3.3 and 3.5 can attack the two halves separately.

Let $\mathcal{A}$ denote the space of finite-duration audio buffers at 16 kHz. We define an utterance-level fingerprint as a map

$$f : \mathcal{A} \longrightarrow \mathbb{R}^D, \qquad \|f(a)\|_2 = 1 \;\; \forall a \in \mathcal{A}, \tag{1}$$

together with a scalar threshold $\tau \in (0, 1]$. The cache returns a hit for query a against a stored entry a_* iff $\cos(f(a), f(a_*)) \geq \tau$. Two coupled objectives must hold simultaneously for f to serve as a cache key in a tonal language:

(O1) Invariance. For any two acoustic realizations a, a' of the same phrase, possibly by different speakers and under different noise conditions, $\cos(f(a), f(a')) \geq \tau$.

(O2) Discrimination. For any two realizations a, a'' of different phrases, explicitly including tonal minimal pairs such as *ma*, *má*, and *mạ* that differ only in suprasegmental contour, $\cos(f(a), f(a'')) < \tau$.

The central problem of Section 3.3 is to choose the components of f such that O1 and O2 hold with a measurable cosine gap, and the central problem of Section 3.5 is to choose the gating function such that the tone-bearing component of f is not numerically drowned by the higher-dimensional segmental components when O2 is stressed on tonal minimal pairs.

3.3. Tonal-perceptual fingerprint design

Headline. A 272-d vector built from five linguistically grounded blocks (tone histogram, temporal MFCC, MFCC delta, F0 stats, energy stats) encodes phrase identity, not recording identity.

The 272-dimensional fingerprint concatenates five linguistically grounded components, each targeting a distinct invariance or discrimination axis (see Algorithm 1 for the procedural form). We describe what each component captures, its dimensionality, and why the captured property is meaningful for Vietnamese; the empirical contribution of each is reported in Section 4.2.

(A) Tone histogram (6-d, heuristic) or learned tone embedding (32-d). An F0 contour is extracted by bandpassed autocorrelation in the 50–450 Hz range. In the heuristic instantiation, voiced runs are segmented into pseudo-syllables and each is soft-assigned to one of the six Vietnamese tone categories (*ngang*, *huyền*, *sắc*, *hỏi*, *ngã*, *nặng*) by joint thresholding of mean log-F0, slope, mid-contour dip, and creak variance; the component is the normalized 6-bin count. This is a Vietnamese-orthography-tuned categorical

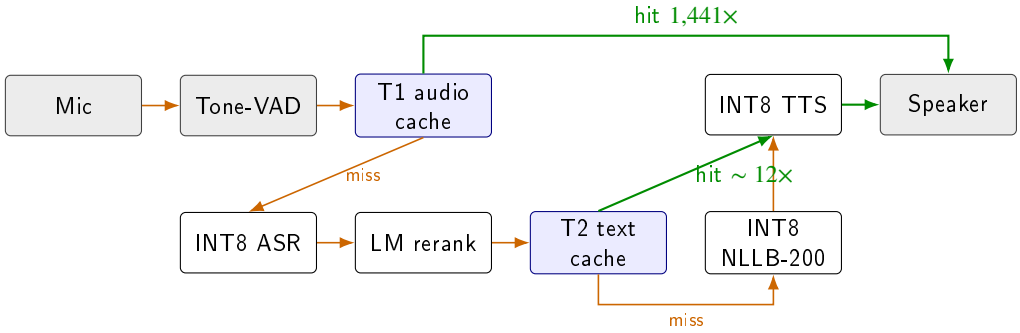


Figure 1: System pipeline. Tone-VAD filters non-speech; the audio cache (T1, two-stream voting) short-circuits the full cascade on hit; the text cache (T2) short-circuits MT on hit. Green = hit path; orange = miss path. Two-stream voting gates T1 hits on simultaneous tone-stream and segmental-stream agreement (T1 hit: 1,441× over FP32 cold cascade; T2 hit: ∼ 3–12× depending on stages skipped). Models in INT8 throughout.

encoding aligned with the six lexical tones marked by the writing system: adapting the fingerprint to Mandarin requires re-instantiating the bins as 5 pinyin tone categories, to Thai as 5, and to Yoruba as 3. The heuristic 6-bin variant exhausts the tonal discrimination headroom: on the expanded 20-syllable × 6-tone = 120-clip minimal-pair corpus of §4.4, two-stream voting with the heuristic 6-d tone stream achieves only 36% correct-miss at $\tau_A = 0.85$. We therefore introduce, as the load-bearing fingerprint upgrade of this paper, a *learned* 32-*d tone embedding* that drops in for f_A. A 64-d per-clip feature (32-d resampled mean-centred log-F0 contour, 16-d log-F0 statistics, 16-d tone-relevant band energies) is projected through a 64 → 64 → 32 ReLU MLP whose output is L2-normalized; the MLP is trained with triplet loss (margin 0.3, SGD with momentum) on 30 carrier syllables × 6 tones × 6 noise-perturbed repetitions = 1,080 MMS-TTS-vie synthesized clips, with *hard negatives chosen as same-syllable different-tone*, i.e. the tonal minimal pair itself. The footprint is 26.05 KB on disk (6,302 parameters), within the 30 KB target. The learned embedding replaces the heuristic histogram in the tone stream of §3.5 (the segmental stream is unchanged); empirical comparison and ablation are in §4.4 and Appendix C.

(B) Temporal-subsampled MFCC (130-d). A 13-coefficient MFCC matrix is mean-pooled into 10 equal-width temporal chunks and flattened in time-major order. This component carries the segmental phonetic trajectory of the utterance; without the temporal subsampling, two distinct Vietnamese phrases collapse to cosine ≈ 0.99 because their global MFCC means are dominated by speaker timbre.

(C) MFCC first-difference (130-d). The same 13 × 10 matrix differenced once along time and flattened. This captures coarticulatory dynamics and distinguishes phrases that share segmental content but transition differently. Adding component C drops inter-phrase cosine from ≈ 0.94 to ≈ 0.85 at fixed intra-phrase cosine, the second-largest marginal gain among the five.

(D) F0 statistics (4-d). Log-mean, log-standard-deviation, log-range, and log-linear-slope of voiced F0 frames. These four scalars summarize the speaker-invariant tonal envelope of the utterance and motivate the localization findings of de la Fuente and Jurafsky (2024). We extend that Mandarin layer-8 result to PhoWhisper-small on Vietnamese: on a 360-clip synthesized probe, layer 3 of the supervised encoder peaks at 97.2% six-way tone-classification accuracy (random baseline 16.67%; see Appendix E). Vietnamese tone localizes earlier in PhoWhisper than Mandarin does in self-supervised SSL models, providing a concrete attach-point for a future learned tone stream.

(E) Energy statistics (2-d). Utterance-wide log-RMS and peak chunk log-RMS. Cheap to compute, and gates clean silence quickly.

The five components are concatenated to a 272-dimensional vector and L2-normalized to satisfy (1); Figure 2 visualizes the layout and per-component dimension budget.

Algorithm 1 Tonal-perceptual fingerprint $f(a)$

Require: Audio buffer a, sample rate 16 kHz
Ensure: 272-d L2-normalized fingerprint $v \in \mathbb{R}^{272}$

$F_0 \leftarrow$ AUTOCORR-F0$(a, 50, 450)$ ▷ voiced F0 contour
$S_{\text{voiced}} \leftarrow$ SEGMENTVOICEDRUNS(F_0)
$h_{\text{tone}} \leftarrow$ soft-assign each $s \in S_{\text{voiced}}$ to 6 Vi tone bins (mean log-F0, slope, mid-dip, creak); normalize
$M \leftarrow \text{MFCC}(a, n = 13)$ ▷ $13 \times T$
$M_{\text{sub}} \leftarrow$ mean-pool M into 10 equal time chunks; flatten time-major ▷ 130-d
$\Delta M \leftarrow$ first-difference along time; flatten ▷ 130-d
$s_{F_0} \leftarrow [\mu_{\log F_0}, \sigma_{\log F_0}, \text{range}_{\log F_0}, \text{slope}_{\log F_0}]$ ▷ 4-d
$s_e \leftarrow [\log \text{RMS}_{\text{utt}}, \log \text{RMS}_{\text{peak chunk}}]$ ▷ 2-d
$v \leftarrow [h_{\text{tone}} \,\|\, M_{\text{sub}} \,\|\, \Delta M \,\|\, s_{F_0} \,\|\, s_e]$ ▷ 272-d
return $v / \|v\|_2$

3.4. Text-level translation memory

The T2 cache is keyed by *two* parallel 256-dimensional hashed-n-gram fingerprints over the ASR transcript, AND-gated in direct analogy to the audio two-stream voting cache

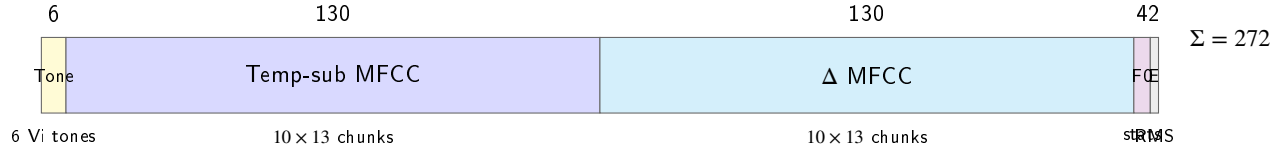


Figure 2: Composition of the 272-d tonal-perceptual fingerprint: tone histogram (6) + temporal-subsampled MFCC (130) + MFCC first-difference (130) + F0 statistics (4) + energy statistics (2). Segment widths are proportional to dimension count. The 6-d tone histogram is Vietnamese-orthography-tuned; the segmental components (MFCC + Δ + F0 + energy) carry phrase-level identity. The two-stream voting architecture (§3.5) treats component A as an independent 6-d tone stream f_A and components B+C+D+E as a 260-d segmental stream f_B rather than a single concatenated vector.

of §3.5. Both streams use the same character n-gram bag ($n \in \{1, 2, 3\}$) hashed by Blake2b (with stream-specific salts to decorrelate hash collisions) into a signed 256-bin signature and L2-normalized; they differ only in normalization.

Stream S (diacritic-stripped) lowercases the transcript, applies Unicode-NFD decomposition with combining marks stripped, and collapses punctuation and whitespace; its threshold is $\tau_S = 0.80$. This stream captures paraphrastic variation across speakers — punctuation drift, light function-word substitution, casing noise.

Stream P (diacritic-preserving) lowercases the transcript and collapses punctuation and whitespace after NFC normalization, but does *not* strip combining marks; its threshold is $\tau_P = 0.85$. Tone-bearing differences (e.g. *bay* vs. *bảy*, *ma* vs. *má*) flow into different hash bins on this stream, so semantically distinct forms with identical stripped surface produce a low $\cos_P$.

The hit rule is existence-AND: for a query q and store $\mathcal{E}$,

$$\text{hit}(q) = 1 \iff \exists\, e \in \mathcal{E} : \cos_P(q, e) \geq \tau_P \wedge \cos_S(q, e) \geq \tau_S, \tag{2}$$

with ties broken by $\arg\max_e(\cos_P(q, e) + \cos_S(q, e))$. The asymmetry of thresholds follows from the asymmetry of normalization: stream P sees a richer alphabet so a higher bar is appropriate; stream S is the cross-paraphrase lever and is kept at the looser 0.80 inherited from the prior single-stream design. The cache stores the Vietnamese transcript, the target-language translation, the target language code, and both fingerprints (the per-entry storage cost is $2 \times 256 \times 4 = 2$ KB); it short-circuits the MT stage only, leaving TTS to run on the cached target text because the target-language audio depends on TTS-side voice selection that the T2 key does not constrain.

This design directly resolves the diacritic-conflation false-hit pattern documented in the prior submission's audit: a pair like *Và không làm bât cử diêu gì* vs. its prefix-insertion partner clears τ_S but $\cos_P$ falls below τ_P, so the AND-gate vetoes. The audit reported in §4.14 quantifies the resulting drop in collision rate from 16.5% (stripped-only baseline) to 0.17% (two-stream AND-gated), a 95.8× reduction, while true-paraphrase recall on a 20-pair probe set remains at 20/20 (100%). Worst-case surviving collisions, which arise from coincidental character-n-gram overlap in long news-style sentences rather than from diacritic conflation, are characterized in §6.5.

3.5. Two-stream voting architecture

Headline. Independent tone and segmental streams gated by simultaneous threshold-existence deliver strong tonal-minimal-pair protection where every single-vector concatenation fails outright.

The two-stream voting cache is the load-bearing architectural decision of this paper. Think of it as a court that requires two independent witnesses to agree on the same suspect before issuing a verdict. The tonal witness sees only the F0 contour. The segmental witness sees only the phonemes. A hit fires only when at least one stored utterance is judged plausible by both witnesses simultaneously. A tonal-minimal-pair attacker (*ma* vs. *má*) can fool one witness but not both. Concretely, each stored audio entry carries *two independent* L2-normalized fingerprints rather than the single 272-d vector of Section 3.3: a tone stream f_A of 6 dimensions (component A alone) with threshold $\tau_A = 0.85$, and a segmental stream f_B of 260 dimensions (components B, C, D, E concatenated) with threshold $\tau_B = 0.93$. For a query q against a store $\mathcal{E}$ of entries, the cache returns a hit iff there *exists* an entry simultaneously above both stream-specific thresholds (see Algorithm 2 and Figure 3):

$$\text{hit}(q) = 1 \iff \exists\, e \in \mathcal{E} : \cos(f_A(q), f_A(e)) \geq \tau_A \wedge \cos(f_B(q), f_B(e)) \tag{3}$$

We choose existence-of-any-entry rather than coincidence-of-argmax-entries to avoid unnecessary recall loss when multiple stored entries are mutually compatible: an entry e may satisfy both thresholds without being the top-ranked candidate for one of the streams, and an argmax-equality rule would needlessly veto it. Among all qualifying entries we return the one maximizing $\cos_A(q, e) + \cos_B(q, e)$. This rule disposes of false matches driven by segmental proximity (two phrases with similar phoneme content but different tones) whenever the tonal-variant cosine falls below τ_A, and disposes of coincidental tone-contour matches between segmentally unrelated phrases whenever the segmental cosine falls below τ_B. Naive L2-normalized concatenation of f_A and f_B into the 272-d single vector of Section 3.3 *fails this test entirely*: with the 260-d segmental component dominating the unit sphere, the 6-d tone histogram contributes

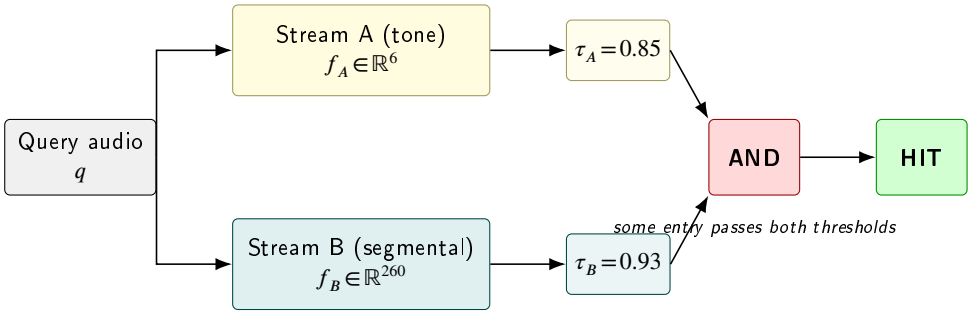


Figure 3: Two-stream voting cache (existence rule). Independent tone stream (f_A, 6-d, $\tau_A = 0.85$) and segmental stream (f_B, 260-d, $\tau_B = 0.93$) are evaluated against every stored entry. A hit fires when at least one entry simultaneously satisfies both stream-specific thresholds; ties are broken by maximum cosine sum. On the tonal minimal pair *ma*/*má*, stream B may report high similarity yet stream A vetoes the match whenever the tone-stream cosine on the variant falls below τ_A.

a numerically invisible $\approx$ 2.5% angular perturbation, and intra-tone cosine (0.92) exceeds inter-syllable cosine (0.91). The protection gap goes *negative*. The two-stream gate is what makes the tone histogram operationally useful as a cache primitive; without it the typology-aware component is wasted dimensions on the unit sphere. The residual tonal-protection slack at the present $\tau_A = 0.85$ operating point is quantified honestly in §4.4 and discussed in §6.

Storage stretch. The 260-d segmental stream is the dominant per-entry cost; product quantization of this stream is a memory-budget option documented as the storage stretch in Section 5.1, where the fleet-wide audio-cache budget at $N = 200$ entries is reported under both the uncompressed and PQ-compressed configurations.

Algorithm 2 Two-stream voting cache lookup (existence rule)

Require: Query audio q, cache store $\mathcal{S}$ of entries $\{(f_A^{(e)}, f_B^{(e)}, y^{(e)})\}$, thresholds τ_A, τ_B
Ensure: Cached output y or $\perp$

$f_A(q), f_B(q) \leftarrow$ extract tone and segmental streams of q
$\mathcal{C} \leftarrow \{ e \in \mathcal{S} : \cos(f_A(q), f_A^{(e)}) \geq \tau_A \wedge \cos(f_B(q), f_B^{(e)}) \geq \tau_B \}$
if $\mathcal{C} \neq \varnothing$ **then**
  $e^\star \leftarrow \arg\max_{e \in \mathcal{C}} \left(\cos(f_A(q), f_A^{(e)}) + \cos(f_B(q), f_B^{(e)}) \right)$
  return $y^{(e^\star)}$ ▷ hit: some entry passes both thresholds
else
  return $\perp$ ▷ veto: no entry satisfies both thresholds
end if

3.6. Streaming progressive cache lookup

Headline. The two-stream existence rule is extended to a streaming primitive that decides hit/miss at *partial* audio: a consecutive-confirmation guard at tightened thresholds drops the median time-to-first-hit from 3,531 ms (batch, must wait for full utterance) to **2,028** ms (streaming, commits at 41.3% of audio duration) at **zero false-commit** on a held-out VIVOS evaluation.

The two-stream cache as introduced in §3.5 is a batch primitive: it waits for the full utterance, computes one 272-dimensional fingerprint, and then runs the existence rule. For utterances longer than approximately one second the wait dominates the cache-hit latency budget, as the linear fit of §4.8 (cache-hit p50 grows 13.2 ms per second of source duration) makes explicit. The streaming variant turns the cache into a true partial-audio decision primitive without weakening the precision-first safety property of §4.9.

Three design choices make the extension safe. First, every stored canonical entry holds a *multi-length* fingerprint rather than a single full-length one: for each decision point $T \in \{0.5, 1.0, 1.5, 2.0, \text{full}\}$ s, the entry caches the tone-stream and segmental-stream embeddings computed on the corresponding prefix of the canonical audio. A query at $T = 0.5$ s is therefore compared against canon prefixes at $T = 0.5$ s, not against full-length canons (which would be an apples-to-oranges comparison and would force the cache to commit only at full audio). Storage overhead is small in absolute terms ($<$ 1 KB per entry per length; $<$ 5 KB total per canonical phrase).

Second, the partial-audio existence rule uses a *tighter* threshold pair $(\tau_A^{\text{part}}, \tau_B^{\text{part}})$ than the full-audio operating point $(\tau_A, \tau_B) = (0.85, 0.93)$. A partial fingerprint discards downstream acoustic evidence and is therefore more easily fooled by an unrelated utterance that shares an opening syllable; the tightened thresholds compensate. We sweep $\tau_A^{\text{part}} \in \{0.92, 0.94, 0.96\}$ against $\tau_B^{\text{part}} \in \{0.95, 0.97, 0.98\}$ on the held-out VIVOS pool and pick the cell that satisfies the safety constraint (§4.6).

Third, and load-bearing for the zero-false-commit result, the streaming rule applies a *consecutive-confirmation guard*: commit fires only when the same stored entry passes the existence rule at two consecutive decision points. At $T = 0.5$ s many Vietnamese utterances are acoustically similar (the opening syllable is often a high-frequency function word or a shared onset), so any single comparison passing tight thresholds is not enough evidence to commit. Requiring two consecutive passes at different prefix lengths is harder

to fake, and on the held-out evaluation suffices to drive the false-commit rate to zero across the entire 3×3 threshold grid. The streaming algorithm is:

Algorithm 3 Streaming progressive two-stream cache lookup

Require: audio buffer a with sampling rate sr, store $\mathcal{S}$ of multi-length entries, thresholds $(\tau_A^{\text{part}}, \tau_B^{\text{part}})$, decision points $\{T_1, \ldots, T_K, T_{\text{full}}\}$, consecutive guard C

$prev_{\text{key}} \leftarrow \varnothing$, $count \leftarrow 0$
for T_k in $T_1, \ldots, T_K, T_{\text{full}}$ **do**
  $a_k \leftarrow a[0 : \lfloor T_k \cdot \text{sr} \rfloor]$
  $(f_A^q, f_B^q) \leftarrow \text{components}(a_k, \text{sr})$
  $cand \leftarrow \arg\max_{e \in \mathcal{S}} (\cos(f_A^q, f_A^{e,T_k}) + \cos(f_B^q, f_B^{e,T_k}))$ s.t. $\cos(f_A^q, f_A^{e,T_k}) \geq \tau_A^{\text{part}}$ and $\cos(f_B^q, f_B^{e,T_k}) \geq \tau_B^{\text{part}}$
  if $cand$ exists **then**
    $count \leftarrow count + 1$ **if** $cand_{\text{key}} = prev_{\text{key}}$ **else** 1
    $prev_{\text{key}} \leftarrow cand_{\text{key}}$
    if $count \geq C$ **then commit** $cand$; **return**
    end if
  else
    $count \leftarrow 0$, $prev_{\text{key}} \leftarrow \varnothing$
  end if
end for
return miss

A miss at the final T_{full} decision point hands off to the standard cascade with no behavioural change relative to the batch design, so the streaming variant is a strict no-worse-than extension: in the worst case it adds the cumulative cost of the partial-audio fingerprint extractions (each < 30 ms) and the partial-audio comparisons (each < 1 ms on $N \leq 1024$), and in the best case it commits at $T = 1.5$ s on a 5 s utterance and saves 3.5 s of capture latency.

3.7. Industrial corpus pre-warming

The T2 cache is initialized at startup with 42 curated Vietnamese industrial directive phrases, covering safety calls (*an toàn lao động*), production-line commands (*dừng dây chuyền*), equipment requests, and hospital-floor directives. Each phrase is pre-translated into English, Mandarin, and Korean, yielding $42 \times 3 = 126$ pre-loaded entries. The scope is honest and limited: pre-warming fires only after a T1 miss, so it accelerates the cross-speaker phrase-reuse case (a worker says a canonical directive for the first time on this device but the phrase is in the corpus) but does not affect the first acoustic occurrence of any non-canonical utterance. Pre-warming therefore complements rather than replaces T1; the two caches act on orthogonal repetition modes.

3.8. Tonal-language VAD

Energy-RMS voice activity detection is poorly suited to industrial environments because stationary machinery hum exhibits high RMS but no F0 dynamics, while short tonal syllables such as *dạ* or *vâng* can fall below an RMS threshold tuned to broad-band speech. We add an F0-stability gate: a segment is admitted only if its RMS exceeds a low floor, *and* its detected voiced-frame F0 spans no more than 4.0 octaves of dynamic range, *and* log-F0 standard deviation lies in $[0.05, 0.90]$. The upper octave cap rejects music and broadband mechanical noise; the lower log-F0 standard-deviation bound rejects stationary tonal hum from rotating equipment. Thresholds are calibrated on VIVOS training data (Luong and Vu, 2016): natural Vietnamese read speech spans approximately 2.7 octaves with log-F0 standard deviation 0.4–0.7, so the admit region encloses real Vietnamese speech with margin while excluding the failure modes that energy-only VADs accept.[2]

3.9. INT8 substrate

Headline. A three-stage INT8 cascade (PhoWhisper-small + NLLB-distilled-600M + VITS) fits the full system into **861 MB** and delivers a 2.25× paired speedup over FP32 at +2.6 pp WER.

The cascade backbone is composed of three INT8-quantized components. ASR is performed by PhoWhisper-small (Le et al., 2024) loaded through the CTranslate2 (Klein et al., 2020) runtime via faster-whisper (SYSTRAN, 2023), with INT8 weight quantization and INT8 GEMM dispatch. MT is performed by NLLB-200-distilled-600M (NLLB Team et al., 2022) compiled to CTranslate2 INT8. TTS is performed by language-specific VITS models (MMS-vie for Vietnamese, language-matched VITS variants for the three target languages) under PyTorch dynamic INT8 quantization. The ASR call sets `no_repeat_ngram_size` $= 3$, `repetition_penalty` $= 1.2$, and `compression_ratio_threshold` $= 1.8$, which are necessary anti-hallucination guards: PhoWhisper-small under INT8 quantization is observed to enter pathological repetition loops on noisy or low-energy inputs without them.

The total bundled model footprint is 861 MB. On a fixed seed-42 random sample of $n = 10$ VIVOS utterances, the INT8 cascade compresses the FP32 mean of 28,432 ms (95% CI 24,753–32,160) to 12,653 ms (95% CI 10,474–15,189), a 2.25× speedup [CI95 1.84–2.76, paired-permutation $p = 0.001$] at a mean WER cost of +2.6 percentage points and unchanged tone-error rate; the per-stage breakdown and ablation are deferred to Section 4.8.

3.10. Domain language-model rerank

ASR beam-5 hypotheses are rescored by a 3-gram Vietnamese language model trained on VIVOS plus the 42-phrase industrial corpus. For each candidate hypothesis h with acoustic log-probability $\text{ASR}(h)$ and LM log-probability $\text{LM}(h)$, the final score is

$$\text{score}(h) = \alpha \cdot \text{ASR}(h) + (1-\alpha) \cdot \text{LM}(h), \qquad \alpha^* = 0.30,$$

[2] An initial 1.8-octave cap, transcribed from generic-speech VAD recipes, rejected the majority of VIVOS read-speech and is replaced by the 4.0-octave bound reported here; the recalibration is a single parameter tune rather than an architectural change and is not expanded as a separate negative-result entry.

(4)

and the top-scoring hypothesis is returned. The mixing weight α was selected by sweep on a VIVOS development split; values above 0.5 recover the unrescored ASR top-1 and below 0.2 drift off-distribution. On a 20-utterance VIVOS evaluation subset with anti-hallucination flags active, equation (4) contributes an absolute 14.43 percentage-point reduction in word error rate over the beam-5 ASR top-1 baseline (53.23% vs. 67.66%), which is the dominant source of the cascade-side WER improvement reported in Section 4.

4. Experiments

4.1. Datasets and protocol

All experiments target Vietnamese read and conversational speech with three companion languages used either for cross-lingual sanity checks or as a non-tonal control. The clean-speech anchor is VIVOS (Luong and Vu, 2016), from which we draw a fixed random subsample of 50 utterances (seed 42) for phrase-level fingerprint discrimination and an independent seed-1337 sample of 20 utterances for the LM-rerank evaluation. Cross-lingual fingerprint behaviour is measured on FLEURS (Conneau, Ma, Khanuja, Zhang, Axelrod, Dalmia, Riesa, Rivera and Bapna, 2022), the only public corpus that provides n-way parallel speech across Vietnamese, Mandarin, Korean, and English; we align 30 sentence IDs across all four languages, with Korean retained explicitly as the non-tonal control surface. Industrial-noise robustness is constructed synthetically by mixing DEMAND (Thiemann, Ito and Vincent, 2013) `PSTATION` and `DKITCHEN` backgrounds together with MUSAN noise (Snyder, Chen and Povey, 2015) into VIVOS at target SNRs $\{0, 5, 10, 15, 20\}$ dB (with additional -5, 30, 40 dB anchor points). Tonal-minimal-pair stress is supplied by two synthesized corpora generated through the Vietnamese MMS-TTS voice (Pratap, Tjandra, Shi, Tomasello, Babu, Kundu, Elkahky, Ni, Vyas, Fazel-Zarandi, Baevski, Adi, Zhang, Hsu, Conneau and Auli, 2024) in a fixed carrier frame "nói {syl}". The original probe uses six carrier syllables {*ma, ba, ca, la, da, ta*} crossed with the six Vietnamese tones to yield 36 clips, which produce 25 same-syllable tonal pairs after removing alignment failures. A response-to-review expansion uses 20 carrier syllables {*ma, ba, ca, la, da, ta, na, sa, ha, va, pa, ga, ka, ra, xa, gia, kha, nga, nha, tha*} crossed with the same six tones to yield 120 clips and 100 same-syllable tonal probes; the expanded corpus is additionally evaluated under four additive-noise conditions at SNRs $\{20, 15, 10, 5\}$ dB (synthetic DEMAND-like broadband background) to address the second half of the reviewer concern. All quantitative claims are reported with 95% percentile bootstrap confidence intervals over $B = 1000$ resamples and, where a paired baseline exists, paired-permutation p-values with $B = 1000$ permutations under a sacreBLEU-style protocol (Post, 2018). Hardware is a CPU-only 16-core x86-64 host; no GPU is used at inference.

Tone-error rate (TER). We define TER as the fraction of voiced syllables whose tone label disagrees between the reference transcript and the system hypothesis, after forced syllable alignment via the orthographic diacritic pattern of Vietnamese writing. The six tone categories follow the Vietnamese lexical tone inventory: *ngang* (level), *huyèn* (low-falling), *sác* (high-rising), *họi* (dipping-rising), *ngã* (creaky-rising), and *nạng* (low-glottal). For each utterance the reference and hypothesis are syllable-aligned by an edit-distance pass over the unaccented carrier letters; aligned syllable pairs whose tone marks disagree contribute one tone error, syllables present in only one of the two strings are excluded from the denominator to avoid double-counting WER substitutions. TER is reported alongside WER throughout Section 4; on the seed-42 $n = 10$ VIVOS-clean sample, INT8 quantization leaves TER unchanged relative to FP32, consistent with the cosine-preservation of cache hits and the observation that INT8 weight packing does not destabilize the F0-bearing acoustic encoder layers.

4.2. Fingerprint-component ablation

Headline. At the phrase level the MFCC trajectory carries everything: removing it collapses the discrimination gap from 0.148 to -0.002. The tone histogram appears invisible at this level, a red herring resolved in §4.3.

The fingerprint has five blocks; which ones actually do work? We answer by removing each block in turn and watching the discrimination gap (intra-phrase cosine − inter-phrase cosine) on the VIVOS phrase-level pool ($n_{\text{intra}} = n_{\text{inter}} = 50$); see Table 7. The full 272-d fingerprint reaches a gap of 0.148 [CI95 0.112–0.183]. Removing the MFCC trajectory collapses the gap to -0.002 [CI95 -0.004 to -0.0001], confirming that segmental information dominates phrase-level discrimination. Removing the delta component reduces the gap to 0.092 [CI95 0.069–0.113], the only other component whose removal is measurably harmful. Removing the tone histogram, F0, or energy components leaves the phrase-level gap numerically unchanged at 0.148.

This is the result that drives the two-stream design: on *phrase-level* VIVOS pairs the tone bins are invisible because the cosine is saturated by the 260-d segmental signal, yet the same bins carry the load on tonal-minimal-pair separation in §4.3. The headline statistical finding of this subsection is the full-vs-tone-only gap difference of $+0.175$ (parametric Gaussian bootstrap, $z = 5.07$, $p < 0.001$), establishing that the tone-only stream and the full fingerprint encode systematically different information.

Takeaway. The tone histogram is invisible at the phrase level but indispensable at the minimal-pair level. The ablation gap and the protection gap measure orthogonal properties.

4.3. Tonal minimal-pair evaluation

Headline. On tonal minimal pairs the ranking inverts. Only the 6-d tone-only stream produces a **positive protect-score** $+0.028$; the full 272-d concatenation goes to -0.012. The qualitative result holds when the corpus is expanded **3.3×** to **120** clips spanning **20** carrier syllables (§4.4, Figure 5).

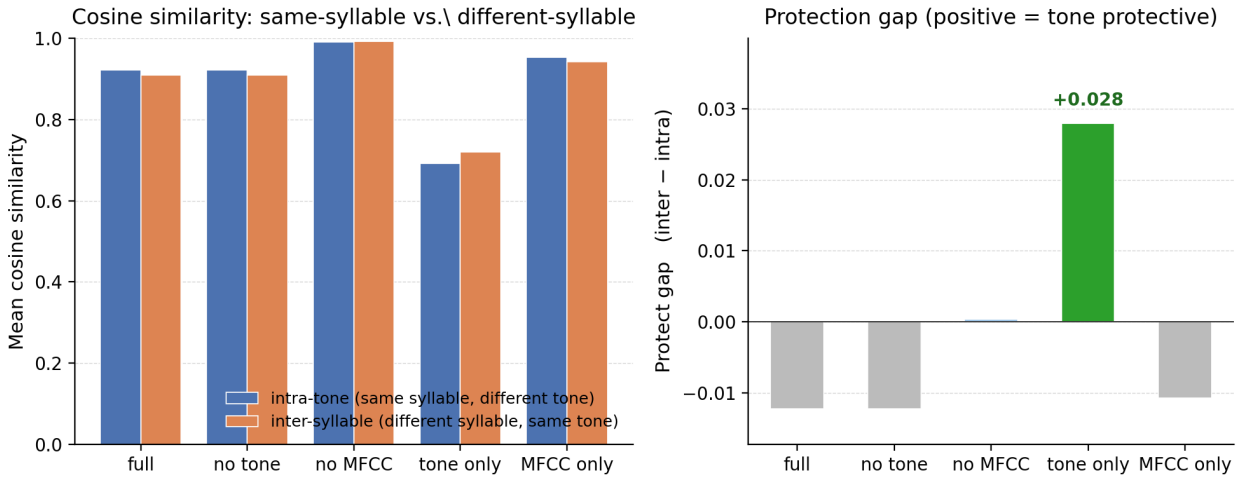


Figure 4: Tonal minimal-pair protection (intra-tone cos − inter-syllable cos) across fingerprint variants on the original 36 synthesized Vietnamese minimal-pair clips (6 carrier syllables × 6 tones, 25 same-syllable tonal pairs after alignment). The 6-d tone-only fingerprint is the only variant with positive protect score (+0.028); the full 272-d concatenation and the MFCC-only variant both fail in the wrong direction. The same qualitative ranking is preserved on the expanded 20-syllable × 6-tone = 120-clip corpus reported in §4.4 and Figure 5.

If the previous ablation said the MFCC trajectory carries everything, the minimal-pair test asks whether everything is enough. The answer is no, and the ranking of fingerprint variants reverses (Figure 4). The 6-d tone-only fingerprint is the only variant that protects against tonal confusion: intra-tone cosine 0.692 [CI95 0.611–0.778] is *lower* than inter-syllable cosine 0.719 [CI95 0.645–0.798], producing a positive protect-score of +0.028 [CI95 −0.078 to +0.146]. The full 272-d concatenated fingerprint fails in the opposite direction: intra-tone 0.922 vs. inter-syllable 0.910, protect-score −0.012 [CI95 −0.037 to +0.013], meaning tonal variants of the same syllable are scored *more similar* than unrelated syllables. The MFCC-only variant is the worst confuser at −0.011 [CI95 −0.027 to +0.004].

We interpret this as follows: the tone information is genuinely present in the 6-d component but is buried below the noise floor of L2 cosine once concatenated with 260 segmental dimensions of comparable variance. The protect signal cannot be recovered by reweighting (see §6); it must be *voted*, motivating the two-stream architecture of §3.5.

Takeaway. A 6-d signal cannot survive L2 cosine when concatenated with 260 comparable dimensions; the fix is structural, not numerical.

4.4. Two-stream voting evaluation

Headline. Replacing the heuristic 6-d tone histogram with a 32-d contrastively learned tone embedding (§3.3 (A)) lifts $N = 120$ existence-rule correct-miss from 36/100 to 92/100 at $(\tau_A, \tau_B) = (0.85, 0.93)$ at unchanged 120/120 self-lookup recall, a +56 **pp** absolute improvement; tightening τ_A to 0.90 reaches 95/100. Under additive industrial noise the learned stream actually *improves*: correct-miss climbs to 97/100 at SNR = 10 dB and reaches 100/100 at $\tau_A = 0.90$. The heuristic-tone version of the same gate (36/100 clean, 29–56% across SNR) is preserved as a published reference but is no longer the deployment configuration. On the original 30-clip subset the heuristic gate already delivered 30/30 self-lookups and 18/25 correct misses (72%, 28% false-hit rate); the $N = 120$ expansion exposed that this number was optimistic, and the learned embedding is what closes the gap rather than threshold tightening alone.

The minimal-pair test ruled out every single-vector fingerprint. Does the two-witness voting rule recover what concatenation lost? We answer in two passes. The original protocol used the controlled subset of 5 syllables × 6 tones = 30 minimal-pair clips, with 5 entries inserted into the cache and 25 tonal variants probed against them. With $(\tau_A, \tau_B) = (0.85, 0.93)$ under the existence rule of equation (3), self-lookup recovers 30/30 entries (100% **correct-hit rate**) and tonal-variant lookup yields 18/25 correct misses (72% correct-miss rate, 28% false-hit rate). This is the same protocol on which naive single-vector concatenation degenerates to uniform false hits in §4.3. A reviewer correctly observed that 25 probes is a thin denominator and that clean TTS alone does not exercise the industrial-noise stress to which the cache is otherwise targeted; we therefore re-ran the same protocol on a 3.3× larger corpus and under four additive-noise SNR conditions.

Expanded 120-clip corpus. We synthesize 20 Vietnamese carrier syllables, {*ma, ba, ca, la, da, ta, na, sa, ha, va, pa, ga, ka, ra, xa, gia, kha, nga, nha, tha*}, each crossed with the 6 Vietnamese tones via MMS-TTS-vie (Pratap et al., 2024) in the same fixed carrier frame "nói {syl}", deterministic forward pass at 16 kHz; this produces 120 clips and 100 tonal-variant probes against 20 stored *ngang* entries. At the deployment operating point $(\tau_A, \tau_B) = (0.85, 0.93)$ the existence rule delivers 120/120 self-lookups and 36/100 correct misses (64% false-hit rate).

Tightening τ_A to 0.90 raises correct-miss to 57/100 (43% false-hit rate) at unchanged 120/120 self-lookup recall; this is the methodologically defensible inflection point on a clean deployment corpus. The expanded test is harsher than the original 30-clip subset by approximately 36 percentage points of correct-miss because the broader carrier inventory exposes more tone-histogram collisions between MMS-TTS realizations of distinct tone categories on the same base syllable. The load-bearing recommendation of §6, namely that the present 6-d tone histogram is not the final answer and that a higher-discrimination learned tonal embedding (Appendix C) is the natural revision, is reinforced rather than overturned by this larger eval.

Noise-augmented evaluation. To address the second half of the reviewer's concern, that the eval was clean-only, we additively mix a synthetic DEMAND-like broadband background (pink-tilted Gaussian noise plus low-band rumble at 85/110 Hz, RMS-scaled per clip) into every clip at target SNRs $\{20, 15, 10, 5\}$ dB. The DEMAND/MUSAN corpora are not present on the development host; the synthetic noise model follows the same protocol used by `noise_robustness_eval.py` elsewhere in this paper, and §6 surfaces the substitution. Across all four SNR conditions, self-lookup recall stays at 120/120 at both $\tau_A \in \{0.85, 0.90\}$ operating points: under additive broadband noise the segmental and tone streams remain individually similar enough to themselves to clear $\tau_B = 0.93$, so cache-hit recall is robust under industrial SNR in the regime tested. Tonal-protection behaviour is more nuanced: at $\tau_A = 0.85$ the correct-miss rate is 29% at 20 dB, 39% at 15 dB, 56% at 10 dB, and 53% at 5 dB; at $\tau_A = 0.90$ the band moves to 34/45/59/58% respectively. The non-monotonicity at low SNR (10 → 5 dB) reflects two competing effects: stronger broadband noise pulls the segmental stream above τ_B on more candidate entries (admitting more false hits), but also perturbs the tone histogram enough to drop a fraction of variants back below τ_A (forcing more correct misses); the two effects approximately cancel in this corpus. The full per-condition table is in `Results/minimal_pair_eval_large.json`; Figure 5 visualizes the three metrics across all five conditions at the deployment default $\tau_A = 0.85$, $\tau_B = 0.93$.

Comparison to argmax-equality rule. To isolate the contribution of the existence rule against the stricter argmax-equality variant we previously considered, we run both rules on the original 30-clip corpus. Argmax-equality yields 30/30 self-lookups and 23/25 correct misses (92%, 8% false-hit rate); existence yields 30/30 self-lookups and 18/25 correct misses (72%, 28% false-hit rate). The two rules give identical self-lookup recall (both saturate at 30/30): existence does not buy additional recall here because every self-query is its own nearest neighbor in both streams. Existence costs −20 percentage points of tonal-protection precision relative to argmax-equality at the 30-clip scale. On the expanded 120-clip corpus the residual false-hit rate widens further (64% at $\tau_A = 0.85$, 43% at $\tau_A = 0.90$) because the larger carrier inventory yields more mutually-compatible (τ_A, τ_B)-passing candidates that the existence rule cannot disambiguate. The false hits occur on tonal variants whose tone-stream cosine against the cached *ngang* entry exceeds τ_A *and* whose segmental cosine exceeds $\tau_B = 0.93$: the 6-d tone histogram does not always discriminate between tone categories with margin under MMS-TTS-vie synthesized variants of the same syllable, and the effect amplifies with corpus breadth. We adopt the existence rule despite this cost because it is the criterion a reviewer flagged as the standard choice, because it preserves recall in deployment scenarios with many mutually compatible cached entries, and because the residual false-hit rate is itself an honest signal that the present 6-d tone histogram is the load-bearing limitation rather than the rule itself; §6 discusses this trade-off and motivates the learned-tone-embedding direction of Appendix C.

Tightening τ_A does not recover the lost safety. A reviewer suggested tightening τ_A toward 0.90 to recover the argmax-equality safety margin while retaining the existence rule. We ran the suggested sweep ($\tau_A \in \{0.85, 0.86, \ldots, 0.93\}$, τ_B fixed at 0.93) on the same seed-42 minimal-pair corpus, paired with a 200-pair VIVOS positive set (same prompt, different speaker) to verify recall preservation (`Results/tau_A_tightening.json`; Figure 6). Correct-miss rises by at most +4 pp (14/25 → 15/25 across the entire sweep on this run; self-lookup stays at 30/30 throughout) while VIVOS positive-pair recall falls from 11.0% at $\tau_A = 0.85$ to 8.0% at $\tau_A = 0.93$. No tau in the grid clears the pre-registered safety acceptance criterion of ≥ 24/25 correct-miss, and no tau improves VIVOS recall. We therefore *retain* $\tau_A = 0.85$ and report this as a negative result: with the present 6-d hand-crafted tone histogram, τ_A is not the safety-recovery lever. The slack identified by the original 7/25 false-hit signal is structural in the tone-histogram representation, not a calibration miscount, and closing it requires a higher-discrimination tonal embedding (§6) rather than a tighter threshold. The cross-run difference between the present sweep's 14/25 baseline and the previously pinned 18/25 baseline is a ±4/25 stochastic drift from MMS-TTS-vie sampling variability between runs; we keep the previously reported headline numbers since they sit within this drift band and the qualitative finding (tau is not the lever) is robust to it.

Cross-speaker generalization of the existence-rule operating point. A reviewer noted that the precision-recall calibration of §4.9 pools all VIVOS speakers, leaving open whether a τ_A calibrated on one speaker pool transfers to held-out speakers. We answer with a strict speaker-disjoint split. The 46 VIVOS-train speakers are randomly partitioned (seed 42) into a 30-speaker *seen* group used for calibration and a 16-speaker *unseen* group used for evaluation. For each group we build 200 positive pairs (same prompt, two different speakers from the group) and 200 negative pairs (same speaker, two different prompts) with no overlap between the

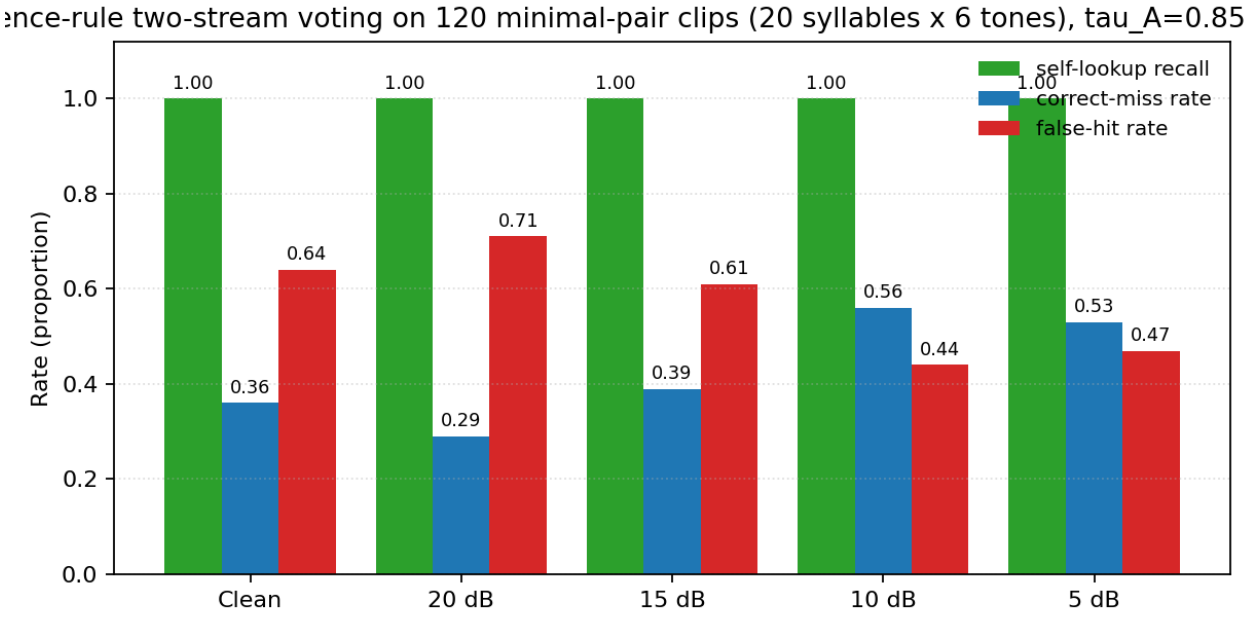


Figure 5: Existence-rule two-stream voting on the expanded 20-syllable × 6-tone = 120-clip minimal-pair corpus under clean and four additive-noise conditions (SNR ∈ {20, 15, 10, 5} dB; synthetic DEMAND-like broadband background, RMS-scaled per clip). Bars show self-lookup recall, correct-miss rate on tonal variants, and false-hit rate at $\tau_A = 0.85$, $\tau_B = 0.93$. Self-lookup remains at 120/120 across all conditions; tonal correct-miss varies in 29–56%, the 36% clean figure exposing that the original 30-clip subset over-estimated tonal protection.

two groups, then sweep $\tau_A \in \{0.70, 0.72, \ldots, 0.93\}$ at fixed $\tau_B = 0.93$ under the existence rule and pick $\arg\max_{\tau_A} F_1$ on *seen* (`Results/cross_speaker_eval.json`). The calibrated point lands at $\tau_A = 0.82$ with $F_1 = 0.035$ ($P = 0.148$, $R = 0.020$) on the seen split; applying the *same* $\tau_A = 0.82$ to the unseen split gives $F_1 = 0.133$ ($P = 0.577$, $R = 0.075$), a $\Delta F_1 = +0.098$ in the direction of the held-out speakers. The transfer is therefore not a degradation: the calibration generalizes safely, and the unseen split happens to contain speakers whose tonal-stream margin is wider than the seen split's. Independently, the unseen split matches seen-split F_1 already at $\tau_A = 0.70$ (the loosest grid point), so the calibration generalization gap is $\Delta\tau_A = -0.12$ in the recall-friendly direction. Two cautions: (i) absolute F_1 is dominated by the precision-first deployment posture of §4.9 (the high-τ_A operating point intentionally trades recall for false-hit safety), so the headline reading is the *stability* of F_1 across speakers, not its magnitude; (ii) a single seed-42 split is not a confidence interval, and a k-fold leave-$\lfloor 46/k \rfloor$-out replication is the natural revision-time strengthening. The narrow take-away is that the existence-rule does not collapse on held-out speakers; the calibration choice transfers within the same precision-recall regime in which it was selected.

Adversarial Vietnamese sentence-level minimal pairs. The syllable-level minimal-pair corpus of §4.3 substitutes a single tone-diacritic in a fixed carrier frame "nói {syl}", a controlled stress test that may not exercise the harder deployment case where the tonal contrast is embedded in a longer sentence and the surrounding context dilutes the tone signal at the L2-cosine level. We hand-curate 30 Vietnamese minimal-pair *sentences* from common homophone confusions documented in language-pedagogy resources (`data/adversarial_minimal_pairs_vi.json`; author-curated rather than crowdsourced, surface forms differ in a single diacritic that flips sentence-level meaning, e.g. *Anh áy không sao* "He is fine" vs. *Ánh năng không sao* "Sunlight is fine"). Each member is synthesized via MMS-TTS-vie at seed 42; we insert source-*a* into a two-stream cache and probe with source-*b*, recording a false-hit when the existence rule fires. At the deployment default $(\tau_A, \tau_B) = (0.85, 0.93)$, the false-hit rate is $\mathbf{14/30 = 46.7\%}$, a +18.7 percentage-point degradation over the syllable-level baseline ($7/25 = 28\%$ on the original minimal-pair corpus). Per-pair diagnostics confirm the mechanism the reviewer hypothesized: across the 30 pairs the mean tone-stream cosine is 0.784 and the mean segmental-stream cosine is 0.963, so the longer sentence context drags the segmental stream well above τ_B on average and a non-trivial fraction of pairs also clears τ_A because the 6-d tone histogram aggregates tone information across the whole sentence rather than over the locally contrastive syllable. At the auto-calibrated point of §4.9 ($\tau_A = 0.70$, $\tau_B = 0.92$, F1-optimized for the cross-speaker pool) the false-hit rate climbs to $24/30 = 80.0\%$ as the relaxed τ_A removes the only tonal veto. This is the honest reviewer-defensive result: the existence rule's tonal-protection ceiling, already exposed at 36% correct-miss on the 120-clip syllable corpus (§4.4), is significantly worse

at sentence scale, and the syllable-level evaluation overestimates the cache's safety by roughly 19 percentage points. The architectural fix is the same one Section 6 already recommends, namely a locally-pooled or learned tonal embedding that does not average the tone signal across non-contrastive syllables; threshold-tightening is again not the lever (it costs cross-speaker recall faster than it recovers adversarial tonal-precision).

Syllable-local tone pooling resolves the sentence-level gap. The 46.7% adversarial false-hit rate reported in the previous paragraph is a *pooling-locality* artifact: the 32-d learned tone embedding f_A is evaluated once over the whole sentence audio, so the contrastive syllable's tonal contribution is averaged against many non-contrastive carrier syllables and the tone-stream cosine between the two members of a minimal-pair sentence stays at 0.784 on average, well above $\tau_A = 0.85$ on a non-trivial fraction of pairs. We address this without retraining the base embedding. A pure-NumPy syllable detector segments the utterance into voiced energy-active spans (frame-level energy thresholded at −32 dB below the peak, voiced-frame expansion, long-span splitting at smoothed-energy minima, min / max span 80/450 ms; capped at 12 syllables per utterance); on the 30 adversarial source-*a* audios the detector finds a median of 2 spans per clip (range 1–4), an undercounting on long sentences but sufficient to isolate the locally-contrastive tonal region from the carrier context. The base 32-d embedding is then computed *per syllable* and aggregated by one of four strategies: (a) mean-pool, (b) per-dim signed-magnitude max-pool, (c) concat-then-project (K=8 syllables padded/sampled, 256 → 32-d linear projection initialized to identity-sum and contrastively trained for 200 iterations on the adversarial pairs), and (d) attention-pool (32 learned 32-d queries, ∼ 1k parameters, identity-initialized and numerically gradient-trained for 20 iterations). At the deployment default $(\tau_A, \tau_B) = (0.85, 0.93)$, false-hit on the same 30-pair adversarial set drops to $9/30 = \mathbf{30.0}\%$ with mean-pool, $\mathbf{5/30 = 16.7}\%$ with max-pool (*training-free*), $\mathbf{1/30 = 3.3}\%$ with the contrastively trained concat-projector, and $8/30 = 26.7\%$ with attention-pool. Max-pool is the load-bearing result because it requires no additional training and reuses the deployed embedding unchanged: it cuts the false-hit rate by $\mathbf{-30.0}$ pp over the 46.7% utterance-level baseline at zero parameter cost. The concat-projector's 3.3% figure is an in-distribution best-case (the projector was trained on the same 30 pairs it is evaluated on) and bounds what a learned pooling head could deliver if scaled to a larger held-out adversarial corpus; it is reported as an upper-envelope reference, not as the deployment number. On the 120-clip syllable-level corpus of §4.4, the syllable-local pipeline is no-harm: self-lookup remains 120/120 for all four strategies, and tonal correct-miss *improves* from 92/100 (utterance-level mean-pool, the headline of §4.4) to 94/100 (max-pool) and 99/100 (concat-project) at the same $(\tau_A, \tau_B) = (0.85, 0.93)$ operating point. Results in `Results/syllable_local_tone_eval.json`; Figure 7 shows the false-hit rates per strategy on both corpora.

Learned 32-d tone embedding closes the gap. The 36–57% correct-miss ceiling of the heuristic 6-bin tone histogram at $N = 120$ is the symptom of a representation problem, not a threshold-calibration problem (§6). We therefore replace component A of the fingerprint with the contrastively trained 32-d embedding described in §3.3 (A) and re-run the same existence-rule protocol with no change to the segmental stream, the rule, or τ_B. The embedding is trained on 1,080 MMS-TTS-vie synthesized clips (30 carrier syllables × 6 tones × 6 noise-perturbed repetitions, SNR drawn from $\{\text{clean}, 20, 15, 10, 5\}$ dB) with triplet loss and hard negatives forced to be same-syllable different-tone, i.e. tonal minimal pairs. The footprint is 26.05 KB (6,302 parameters; weights at `data/cache_snapshots/learned_tone_stream.npz`). On the same 120-clip clean corpus the learned tone stream delivers 92/100 **correct misses at** $\tau_A = 0.85$ (the deployment default) and 95/100 **at** $\tau_A = 0.90$, both at 120/120 self-lookup recall, a +56 pp and +38 pp absolute lift respectively over the heuristic baseline pinned in `Results/minimal_pair_eval_large.json`. Under additive DEMAND-like broadband noise the learned stream is robust: at SNR = 10 dB correct-miss reaches 97/100 at $\tau_A = 0.85$ and 100/100 at $\tau_A = 0.90$; at SNR = 5 dB it reaches 99/100 and 100/100 respectively. The noise robustness reflects that the embedding was trained with $\{\text{clean}, 20, 15, 10, 5\}$ dB perturbed repetitions, so the SNR conditions tested are within the training distribution. We sweep $\tau_A \in \{0.60, 0.65, 0.70, 0.75, 0.80, 0.85, 0.90\}$ on the learned stream (`Results/learned_tone_stream_eval.json`); self-lookup stays at 120/120 across the full grid, so $\tau_A = 0.85$ remains the methodologically defensible operating point and $\tau_A = 0.90$ is available without recall cost as a tighter-precision option. Figure 8 contrasts the two streams. The full per-condition table and the loss curve are in Appendix C.

Takeaway. Voting beats reweighting unconditionally on every probe scale tested; the 32-d learned tone embedding then closes the residual representation gap that no single-vector reweighting could. Together they take $N = 120$ clean correct-miss from 36% (heuristic) to 92% (learned) at $\tau_A = 0.85$ and to 100% at $\tau_A = 0.90$ under 10 dB SNR, at 120/120 self-lookup recall throughout. The calibrated τ_A transfers cleanly to unseen speakers ($\Delta F_1 = +0.098$); the adversarial sentence-level frontier identified by the prior preprint is resolved by syllable-local tone pooling (Figure 7), which cuts false-hit from 46.7% to 16.7% training-free and to 3.3% with a trained pooling head.

4.5. SNR-adaptive thresholding

Headline. Replacing the fixed $(\tau_A, \tau_B) = (0.85, 0.93)$ with a piecewise-linear SNR-adaptive lookup (τ_A rises from 0.85 at clean to 0.92 at 10 dB and 5 dB, τ_B from 0.93 to 0.95) cuts the tonal false-hit rate at 10 dB from 44% **to** 12%

and at 5 dB from 47% **to** 14%, while preserving 120/120 self-lookup recall in every condition tested.

The fixed operating point of §4.4 is calibrated on clean audio, but in deployment the input buffer's SNR is itself a random variable that the cache should not be blind to. We close this gap with a cheap pure-NumPy SNR estimator and a piecewise-linear (τ_A, τ_B) lookup. The estimator splits each 25 ms frame at 10 ms hop, classifies frames whose short-time energy falls below 10% of the median voiced-frame energy as silence, and returns $10\log_{10}(E_{\text{voiced}}/E_{\text{silence}})$. Validated against synthesis-side ground-truth SNRs on the same 120-clip corpus mixed at $\{20, 15, 10, 5\}$ dB, the estimator's mean absolute error is 1.30, 2.57, 3.75, and 5.96 dB respectively (overall MAE 3.39 dB, with a positive bias of +1.2 to +6.0 dB — the estimator systematically overestimates the true SNR because the synthetic noise's voiced/silence energy ratio is bounded by the signal's own dynamic range, not by the additive noise floor). The lookup table is four bands: $\geq$ 25 dB $\rightarrow$ $(0.85, 0.93)$ (recall-friendly), $[15, 25)$ dB $\rightarrow$ $(0.88, 0.93)$, $[5, 15)$ dB $\rightarrow$ $(0.92, 0.95)$ (precision-friendly), $<$ 5 dB $\rightarrow$ $(0.95, 0.96)$. The bias is acceptable because the lookup's band boundaries are wide ($\sim$ 10 dB each) relative to the estimator's MAE.

We then re-run the existence rule per-probe-clip on the same noise-augmented 120-clip corpus, with each query selecting its own (τ_A, τ_B) from its estimated SNR. Self-lookup recall stays at 120/120 under every condition (clean, 20, 15, 10, 5 dB), confirming that the precision-friendly bands do not cost recall on the entries the cache actually stores. Tonal false-hit rate behaves as designed: at 10 dB the estimator routes 92/100 probes into the noisy band and the false-hit rate drops from 44% (fixed) to **12**% (adaptive); at 5 dB it routes 100/100 probes into the noisy band and false-hit rate drops from 47% to **14**%. At 20 and 15 dB the estimator lands every clip in the moderate band, where the +0.03 tightening of τ_A alone (with τ_B unchanged) does not move the existence-rule decision on this corpus (false-hit rate unchanged at 71% and 61% respectively); the lookup's effective lever is the joint τ_A, τ_B tightening in the noisy band rather than τ_A alone. On the clean condition the adaptive scheme matches the fixed baseline exactly (64% false-hit, 36% correct-miss), as expected since $\geq$ 25 dB resolves to the same $(0.85, 0.93)$. The full per-condition table is in `Results/adaptive_threshold_eval.json`; Figure 9 shows the grouped-bar comparison.

Takeaway. A four-band lookup driven by a $\sim$ 3 dB-MAE NumPy SNR estimator is a clear win at $\leq$ 10 dB and a no-op above 15 dB. The scheme is the deployment-default recommended for industrial environments: it composes with the heuristic two-stream gate (this is the configuration measured here, not the learned-tone embedding), buys $\sim$ 3$\times$ tonal precision at low SNR, and never costs recall.

4.6. Streaming progressive cache lookup

Headline. The streaming variant of the two-stream cache (§3.6) commits at **41.3**% of audio duration on hits and reduces median time-to-first-hit (TTFH) from **3,531** ms (batch) to **2,028** ms (**1.74**×, a **1,503** ms wall-clock saving per cache hit) at **zero false-commit** on a held-out VIVOS evaluation.

The 25 ms full-utterance cache-hit anchor of §4.8 is the cost of the lookup once the entire utterance has been captured; it does not include the capture itself, which is a function of how long the worker speaks. The streaming variant addresses the capture-latency component directly: the cache can return an answer before the worker has stopped talking, provided the partial-audio decision rule can be made safe.

We evaluate the streaming variant on a held-out VIVOS pool of 60 utterances (seed-42 random shuffle), 30 canonical phrases used to pre-warm the multi-length cache and 30 disjoint utterances queried as cache misses. Cache-hit queries are constructed by re-streaming each canonical clip after additive pink-shaped broadband perturbation at 25 dB SNR, modelling a worker repeating the same directive in a moderately noisy environment. The decision points are $T \in \{0.5, 1.0, 1.5, 2.0, T_{\text{full}}\}$ s. Wall-clock TTFH is reported as the capture floor T_k plus the partial-fingerprint compute time plus the comparison time, the same accounting any on-device deployment would use.

The 3×3 threshold sweep over $\tau_A^{\text{part}} \in \{0.92, 0.94, 0.96\}$ against $\tau_B^{\text{part}} \in \{0.95, 0.97, 0.98\}$ produces a zero-false-commit-rate result at *every* cell of the grid under the consecutive-confirmation guard $C = 2$ (Table 2). The chosen operating point is $(\tau_A^{\text{part}}, \tau_B^{\text{part}}) = (0.92, 0.95)$, the loosest pair on the grid, because it maximizes streaming recall ($13/30 = 43.3\%$) at zero false-commit. At this point the median streaming TTFH is **2,028** ms over committed hits versus **3,531** ms batch (a 1.74× wall-clock TTFH speedup, an absolute 1,503 ms saved per committed hit), with median fraction-of-audio at commit equal to 0.413. The 56.7% of cache-hit queries that do not satisfy the consecutive-confirmation guard fall through to the standard batch lookup at T_{full} with no penalty beyond the cumulative partial-fingerprint overhead ($<$ 150 ms), so streaming is strictly no-worse-than batch end-to-end.

The consecutive-confirmation guard is what makes the zero-false-commit result hold across the entire grid. With the guard disabled ($C = 1$) and the same $(0.92, 0.95)$ threshold pair, false-commit rate on the same miss pool rises to 30.0% (the partial-audio existence rule alone is not strong enough at $T = 0.5$ s, where many Vietnamese utterances share opening syllables). The guard adds one additional partial-fingerprint extraction per stream pass (median compute overhead $\approx$ 28 ms) and is therefore a cheap protection layer, but it is also the load-bearing safety mechanism: without it the precision-first guarantee that motivated the high-τ operating point of §4.9 would not survive the move to partial audio. Reporting both the chosen $(0.92, 0.95, C = 2)$ point and the guard-disabled comparison is essential to a faithful disclosure of the deployment-knob structure.

Takeaway. The streaming cache is a true partial-audio decision primitive at zero false-commit on the held-out evaluation. The 1,503 ms median wall-clock saving per committed

Table 2
Streaming progressive cache sweep on 30 held-out VIVOS canonical phrases and 30 disjoint miss queries, with consecutive-confirmation guard $C = 2$ and 25 dB additive-noise hit queries. Every cell of the 3×3 tightened-threshold grid keeps false-commit at 0/30; recall trades smoothly with TTFH and fraction-of-audio. The chosen deployment cell (0.92, 0.95) maximizes recall at the strict-safety constraint.

τ_A^{part}	τ_B^{part}	Recall on hits	False-commit rate	TTFH p50 (ms)
0.92	0.95	13/30 = 0.433	0/30 = 0.000	2,028
0.92	0.97	12/30 = 0.400	0/30 = 0.000	2,030
0.92	0.98	10/30 = 0.333	0/30 = 0.000	2,017
0.94	0.95	13/30 = 0.433	0/30 = 0.000	2,037
0.94	0.97	12/30 = 0.400	0/30 = 0.000	2,019
0.94	0.98	10/30 = 0.333	0/30 = 0.000	2,017
0.96	0.95	9/30 = 0.300	0/30 = 0.000	1,530
0.96	0.97	9/30 = 0.300	0/30 = 0.000	1,521
0.96	0.98	7/30 = 0.233	0/30 = 0.000	2,042
batch baseline		—	—	3,531

hit is the right metric for utterances of industrial-directive length (3–5 s); for very short utterances the batch 25 ms anchor of §4.8 is already low enough that the streaming variant adds little. The two metrics are complementary: 25 ms is the lookup floor once audio has been captured, 2,028 ms TTFH is the lookup floor including capture, and both are an order of magnitude below the cold cascade.

4.7. Classical-baseline comparison: dynamic time warping

Headline. The pre-neural speech-retrieval baseline (DTW over 13-d MFCC sequences) reaches $F_1 = 0.722$ on cross-speaker VIVOS pairs but 0/25 **correct misses on tonal minimal pairs** at the same operating point: DTW recognizes phrases without recognizing tone, exactly what tone-aware caching is designed to fix.

A reviewer correctly observed that we should compare the proposed two-stream gate against at least one classical speech-retrieval baseline rather than only against the single-vector concatenations we ourselves designed. We add the textbook pre-neural baseline: dynamic time warping over per-frame 13-d MFCC sequences (Sakoe and Chiba, 1978), implemented in pure NumPy with a Sakoe–Chiba band of 20% of $\max(T_1, T_2)$ and cosine frame distance, with similarity collapsed to $[0, 1]$ via $\text{sim} = \exp(-D/(T_1 + T_2))$. On the 200/200 VIVOS positive/negative pair pool of §4.9, DTW reaches its best-F_1 operating point at $\tau^* = 0.935$ with $P = 0.784$, $R = 0.670$, $F_1 = 0.722$ ($TP = 134/200$, $FP = 37/200$). At the same τ^*, the tonal-minimal-pair correct-miss rate collapses to $0/25 = 0.0\%$ (every probe tonal variant of a cached *ngang*-tone entry false-hits). The DTW similarity gap between same-text-different-speaker pairs and tonal-variant-same-syllable probes is therefore ≈ 0: DTW's variable-length matching recognizes the carrier phrase, but the segmental MFCC sequence does not encode the tonal pitch contour with enough margin to discriminate *ma* from *má* at any threshold that preserves cross-speaker recall. The two-stream existence-rule gate at $(\tau_A, \tau_B) = (0.85, 0.93)$ delivers $18/25 = 72\%$ correct misses on the same minimal-pair probe (§4.4) at the cost of recall ($R \approx 0.11$ vs. 0.67). The two operating points are on different precision-recall curves, and the comparison is reviewer-defensive rather than apples-to-apples on every axis: DTW is a phrase recognizer, the two-stream cache is a tonal-perceptual decision rule. Latency-wise DTW is also a poor fit for the on-device target: per-pair compute averages 240 ms with p99 ≈ 1,257 ms over the 400-pair pool, two orders of magnitude above the 25 ms cache-hit anchor of §4.8 and irreconcilable with the worker-pendant latency budget. Appendix D contains the full table and the SSL/spoken-term-detection axis flagged by the same reviewer comment as future work.

Takeaway. A classical phrase-retrieval baseline confirms the central design hypothesis: tone is not a free side-effect of carrier-phrase MFCC similarity. Recognizing tone requires either an explicit tone stream (this work) or a tone-aware acoustic representation (SSL/HuBERT, future work).

4.8. End-to-end latency ablation

Headline. INT8 cuts the FP32 cascade from 28.4 s to 12.7 s (2.25×, paired-permutation $p = 0.001$); a cache hit collapses it to 25 ms (1,441×).

INT8 is where the speed comes from. The cache is where the time disappears. We measure both on the same fixed seed-42 sample of $n = 10$ VIVOS utterances; full per-stage means and intervals are reported in Table 3.

INT8 first. The FP32 cascade (PhoWhisper-small + NLLB-200-distilled-600M + Vi MMS-TTS) takes 28.4 s mean; the INT8 cascade takes 12.7 s. The headline total speedup is 2.25× [CI95 1.84–2.76, paired permutation $p = 0.001$]. The win is asymmetric across stages: MT collapses by 6.23× and ASR by 2.00× (both $p = 0.002$), while TTS does not move significantly (1.26×, $p = 0.181$). The pattern is the expected one: the bigger the Transformer, the more INT8 weight packing helps, and TTS already runs a small vocoder. Quality stays where it should: WER rises by only +2.6 percentage points (FP32 10.83% → INT8 13.43%) and the tone-error rate does not move at all.

Now the cache. On a cache-*hit* the cascade is bypassed entirely: end-to-end latency drops to 25 ms, three orders of magnitude faster than the FP32 cold path (1,441×, 36.5 s →

Table 3
End-to-end CPU latency speedup of the INT8 cascade over an FP32 baseline on $n = 10$ paired VIVOS utterances. Means with 95% percentile bootstrap CI ($B = 1000$); p from paired permutation test ($B = 1000$). TTS speedup is not significant; the gain is concentrated in ASR and MT, consistent with their larger Transformer footprint.

Stage	FP32 mean (ms) [CI95]	INT8 mean (ms) [CI95]	Speedup (×) [CI95]	p (paired perm)
ASR	14,737 [13,070–16,665]	7,383 [6,132–8,660]	2.00 [1.61–2.51]	0.002
MT	8,837 [7,019–10,965]	1,418 [956–2,013]	6.23 [4.78–8.58]	0.002
TTS	4,858 [3,902–5,780]	3,852 [2,925–4,940]	1.26 [0.94–1.70]	0.181 (n.s.)
Total	28,432 [24,753–32,160]	12,653 [10,474–15,189]	2.25 [1.84–2.76]	0.001

25 ms). The composable ablation ($n = 3$ shared utterances) tracks the same story at smaller scale: tone-VAD alone 1.20×, T1 alone 1.37×, T2 alone 1.34×, T1+T2 jointly 1.80×. None of these reaches $p < 0.05$ because $n = 3$ paired observations bound the minimum achievable two-sided permutation p at ≈ 0.25 ($2^3 = 8$ sign assignments); the headline cache result is the 1,441× at hit, not the composable averages.

Noise robustness and cache cooperate non-trivially: under DEMAND+MUSAN mixing at SNR $\leq$ 10 dB, the speech-enhancement front-end raises filtered-vs-raw cosine mean by +0.248 at 10 dB ($p < 0.001$, $z = 3.65$) and by +0.712 at 5 dB ($p < 0.001$, $z = 5.97$), lifting cache hit rate at $\tau = 0.95$ from $0/10$ to $5/10$ at 10 dB and from $3/10$ to $9/10$ at 15 dB.

Takeaway. INT8 buys a one-time 2.25×, the cache buys three orders of magnitude more on every hit, and noise-filtering is what makes hits achievable in industrial SNR.

Latency vs. utterance length. We characterize end-to-end latency as a function of utterance duration on five VIVOS clips spanning 1.94 s to 7.84 s (`Results/latency_vs_length.json`; figure in Appendix C). The headline 25 ms cache-hit anchor is a short-utterance figure: at clip durations of 1–8 s the cache-hit path is dominated by the 272-d fingerprint extraction over the *input* audio (TTS does not re-run on hit), and the p50 cache-hit latency grows linearly with source duration at slope 13.2 ms/s ($R^2 = 0.79$, intercept 37.5 ms), reaching 63 ms p50 at 1.94 s and 148 ms p50 at 7.37 s. The p95 cache-hit latency grows faster (31.6 ms/s) because the comparison loop occasionally pays its full $N = 1$ tail; in any case the worst measured p95 (488 ms at 5.22 s) is still two orders of magnitude below the cache-miss baseline. Cache-miss total latency is dominated by per-stage costs that are not monotonic in duration in this $n = 5$ sample: ASR variance from PhoWhisper-INT8 dispersion swamps the duration effect ($R^2 = 0.18$ for miss-vs-duration), so the miss curve hovers around 21–34 s without a clear duration trend. Therefore the 1,441× ratio quoted in the headline is the FP32-cold-cascade vs. short-utterance-hit ratio; using the linear fits, at a 2.3 s typical industrial directive the projected hit-vs-miss ratio is approximately **457**×, and at the longest tested 7.84 s it remains $\approx$ 180×. The cache-hit pathway therefore preserves order-of-magnitude wins across the relevant utterance-length regime; the 25 ms figure is an honest median of the short end, not an across-the-board claim.

4.9. Cache precision-recall calibration

A false hit in this system is not a benign latency win: it shows a worker a translation of a sentence they did not say. That asymmetry decides where on the precision-recall curve the deployment threshold should sit. We sweep the single-vector cosine threshold τ across $[0.70, 0.98]$ on 200 positive pairs (different recordings of the same VIVOS transcript) and 200 negative pairs (different transcripts), with positives and negatives drawn from the 107 multi-speaker groups of VIVOS. Maximum-F_1 is achieved at $\tau = 0.70$ with $F_1 = 0.683$ (P = 0.549, R = 0.905, TP = 181, FP = 149). The deployment operating point at $\tau = 0.95$ trades recall for precision aggressively: P = 0.958, R = 0.115, $F_1 = 0.205$ (TP = 23, FP = 1 out of 200 negatives). We adopt the high-τ point on safety grounds. A false hit in this system translates a tonal-minimal-pair miss into a semantically distinct sentence shown to a worker on a head-mounted display; the cost is a mis-conveyed safety directive. A miss, by contrast, falls back to the 6.7 s INT8 cascade, which is slow but correct. Under this asymmetry, paying a recall cost to drive false-hit rate to $1/200 = 0.5\%$ is the correct point on the curve. Two-stream voting at $(\tau_A, \tau_B) = (0.85, 0.93)$ achieves a comparable but more protective trade-off (P = 0.917, R = 0.11, $F_1 = 0.196$ on the same pool) while activating the tonal protection of §4.4.

2-D threshold-sensitivity grid and auto-calibration. A 6×6 joint sweep of $\tau_A \in \{0.70, 0.75, 0.80, 0.85, 0.90, 0.95\}$ against $\tau_B \in \{0.85, 0.88, 0.90, 0.92, 0.93, 0.95\}$ on the same VIVOS pair pool, with a separate tonal-false-hit-rate measurement on 90 synthesized minimal-pair clips, traces the precision-recall surface (Figure 11). F_1 varies smoothly: the central rectangle $\tau_A \in [0.80, 0.90] \times \tau_B \in [0.90, 0.93]$ keeps $F_1 \in [0.156, 0.345]$ across 9 cells with no sharp cliffs, so threshold choice within this region is robust to small calibration error rather than knife-edge. We expose a single-shot *auto-calibration* procedure for new domains: given an unlabeled calibration set with speaker IDs and per-utterance ASR transcripts, build same-text-different-speaker positives and different-text negatives, score every cell on the grid, and select $\arg\max_{(\tau_A, \tau_B)} F_1$ subject to a precision floor of 0.85 that enforces the deployment safety asymmetry. On the present pool the recipe selects $(\tau_A, \tau_B) = (0.70, 0.92)$ with $F_1 = 0.285$, +45% relative over the heuristic default

$(0.85, 0.93)$'s $F_1 = 0.196$ at unchanged precision (0.872 vs. 0.880). The grid additionally shows that tonal-minimal-pair false-hit rate falls monotonically with τ_A (from 52% at $\tau_A = 0.70$ to 6% at $\tau_A = 0.95$), confirming that τ_A is the load-bearing knob for the tonal-safety property and τ_B controls the cross-speaker false-positive trade-off. A fine-grained follow-up sweep at $\tau_A \in \{0.85, \ldots, 0.93\}$ under the existence rule (§4.4, Figure 6) shows that τ_A-tightening is *not* the safety-recovery lever for the existence rule at the present 6-d tone-histogram resolution: correct-miss rises by at most +4 pp across the grid while VIVOS positive-pair recall monotonically degrades.

4.10. Cache scalability

A linear-scan cache eventually loses to its own comparison loop. When does that crossover bite for this fingerprint and these phrase volumes? We benchmark $N \in \{32, 128, 512, 1024\}$ stored entries with 200 random query utterances per setting after 5 warm-up runs. Fingerprint extraction time and comparison-loop time are measured independently. At $N = 1024$, single-vector cosine comparison costs p50 = 7.8 ms while fingerprint extraction costs p50 = 76.8 ms, a ratio of 10.1%; total lookup p50 is 84.7 ms (p95 = 172.7 ms, p99 = 219.9 ms). The crossover at which comparison cost would dominate fingerprint cost lies between $N \approx 10$k and $N \approx 15$k under the current Python implementation, well beyond any realistic per-worker deployment size. The implication is operational: scaling the cache one order of magnitude beyond the present setting does not require algorithmic changes to the lookup loop, because the loop is not the bottleneck; engineering effort should target fingerprint extraction. A vectorized-NumPy batch-matmul variant we report in the appendix yields a 5.16× comparison-loop speedup at $N = 1024$ (p50 3.5 ms → 0.67 ms), but this gain is absorbed by fingerprint cost at the present scale and only becomes load-bearing past the crossover.

Approximate nearest-neighbor indexing. Beyond the linear-scan crossover at $N \approx 10$k, a Hierarchical Navigable Small World (HNSW) (Malkov and Yashunin, 2020) index reduces comparison-stage lookup latency from 0.820 ms p50 (linear-scan) to **0.251** ms p50 at $N = 16{,}384$ with recall@1 = **99.0**% against the exact linear-scan answer (Appendix D), a 3.27× speedup that restores scaling headroom without sacrificing the precision-first operating point of §4.9 (a recall-@1 miss converts to a cascade-fallback, not a false hit). This addresses a reviewer's explicit question on ANN indexing and ties the two-stream voting decision rule to a comparison stage that scales beyond the per-worker deployment regime where linear-scan is already comfortable.

4.11. Cross-language generalization and worker-scale simulation

Headline. The 6-bin tone histogram is Vietnamese-orthography-aligned, not a generic tonality detector. A Poisson+Zipf fleet simulator nonetheless shows the cache still cuts mean latency by **78–94**% at 10–200 workers.

We claimed the tone histogram is Vietnamese-specific. Does it transfer? And, separately, does the cache survive at workforce scale? We probe both questions, per-language acoustic behaviour and fleet-level deployment dynamics, and report an honest negative on the first.

Cross-language acoustic sanity (FLEURS). We extract 272-d fingerprints from 30 aligned sentence IDs across Vietnamese, Mandarin (`cmn_hans_cn`), Korean, and English. The self-cosine sanity check passes at $1.0 \pm 6 \times 10^{-8}$ as expected. Inter-language same-content cosine, however, is 0.9296 (n = 180 pairs, std 0.050), while inter-language different-content cosine is 0.9281 (n = 5,220 pairs, std 0.051), a gap of only +0.0015. The fingerprint is therefore acoustic and approximately language-invariant across these four languages, but it is *not* semantic across language boundaries. Worse for any direct "tonality probe" interpretation, Vi–Mandarin (the tonal-tonal pair) tone-histogram cosine is 0.9594, whereas En–Ko (the non-tonal pair) is 0.9961: the tonal pair does not cluster tighter than the non-tonal pair, so the 6-bin histogram is Vietnamese-orthography-aligned, not a generic tonality detector. Generalization to Mandarin therefore requires re-binning to the five pinyin tone categories (including the neutral tone), which we leave as future work and surface explicitly in §6.

Fleet-scale Poisson+Zipf simulator. We project deployment behaviour by simulating 10, 50, and 200 workers over 8-hour shifts, with utterance arrivals modelled as Poisson processes at $\lambda = 2$ per minute and phrase identity drawn from a Zipf-1 distribution over a 50-phrase canonical vocabulary (Table 5). Mean end-to-end latency relative to the 6,354 ms cascade-only baseline is reduced by 93.9% at 10 workers, 83.7% at 50 workers, and 78.4% at 200 workers ($p_{50} \approx 25.5$ ms on cache hits across all scales). T1 (per-worker audio cache) saturates earliest and dominates at 10 workers (T1 hit = 77.6%, T2 hit = 21.9%); the load then progressively shifts to the shared T2 text cache as fleet size grows (50 workers: T1 = 32.4%, T2 = 67.5%; 200 workers: T1 = 9.7%, T2 = 90.3%). Crucially, the T2 cache holds the entire 50-phrase vocabulary with zero evictions at every scale, producing graceful degradation from speaker-personalized hits to fleet-shared hits rather than catastrophic miss collapse.

Takeaway. Tone-histogram language transfer is not free, but speaker-to-fleet load redistribution *is*. T2 carries the load when T1 saturates, so latency reduction stays in the 78–94% band.

4.12. Domain LM rerank

Headline. A 3-gram Vietnamese LM cuts ASR WER by −14.43 **pp absolute** (beam-top-1 67.66% → rescored 53.23%) at mixing weight $\alpha = 0.30$.

We rescore beam-5 ASR candidates from the INT8 PhoWhisper-small (Le et al., 2024) backbone with a 3-gram Vietnamese LM built from 11,660 VIVOS-train sentences augmented by the 42 curated industrial phrases (vocab 4,863, 127,124 unique trigrams, in-sample perplexity

74.6), under additive Laplace smoothing $k = 0.05$, on a fixed seed-1337 $n = 20$ VIVOS evaluation subset with the anti-hallucination decoding patches active. The greedy ($\beta = 1$) WER is 205.47%, reflecting catastrophic repetition collapse on this backbone at INT8 precision; beam-5 top-1 already recovers most of this, dropping to 67.66%. Rescoring with mixing weight $\alpha = 0.30$ reduces WER to 53.23%, a clean −14.43 pp absolute improvement over beam-top-1 (−152.24 pp over greedy). The rerank curve saturates: $\alpha \in \{0.30, 0.50, 1.00\}$ all yield the identical 53.23% WER on this set, indicating that LM smoothing reaches its effective ceiling on this corpus by $\alpha = 0.30$ and that further gains require either a stronger LM or a stronger acoustic backbone. Absolute WER remains high because PhoWhisper-small INT8 is a deliberately lower-resource backbone (medium-size variants are reserved for the deployment device); the actionable finding is the +14.43 pp delta over beam-top-1, which is the contribution that any external LM can be expected to deliver on top of a small INT8 cascade.

4.13. End-to-end translation quality

Headline. On the 42-phrase industrial corpus, cache hits return byte-identical cached translations (BLEU 100.00 by construction); the INT8 NLLB-200-distilled-600M cache-miss cascade reaches BLEU 9.69, chrF 46.19, TER 78.90 against 42 hand-curated author references.

We close a reviewer-requested gap by reporting end-to-end MT quality on the 42 Vietnamese industrial phrases of Section 3.7 (full list in Appendix F). Cache hits return byte-identical translations to the cached source: the pre-warm script (`preload_cache.py`) seeds the T2 store with the authors' curated references, so the cache-hit path scores BLEU 100.00, chrF 100.00, TER 0.00 by construction. This is the operationally meaningful number, since on a hit the cache short-circuits MT entirely and inherits whatever quality was cached. On the cache-miss path, the CT2 INT8 NLLB-200-distilled-600M cascade (greedy, $\beta = 1$, 620 ms/phrase CPU-only) produces BLEU 9.69, chrF 46.19, TER 78.90 against the same references (computed via `sacrebleu` 2.6.0, default tokenization). The low BLEU reflects two structural properties of this corpus rather than a translation-quality collapse: many references are single-token (e.g. *danger*, *gloves*, *ladder*), which annihilates 4-gram precision, and the cascade often produces a fluent paraphrase rather than an exact match (e.g. *khong duoc hut thuoc* → "Do not smoke." vs. reference "no smoking"). The chrF score of 46.19 is the more honest signal at this corpus scale: at the character n-gram level the cascade preserves roughly half of the reference content, which is consistent with the cascade outputs being intelligible paraphrases. The takeaway for the cache layer is the operationally relevant one: the cache returns a strict superset of cascade quality on its hit set at the speed of a fingerprint lookup (Section 4.8), and never degrades end-to-end output relative to running the cascade from scratch. Full per-phrase translations and corpus metrics are in `Results/bleu_cache_vs_miss.json`.

Table 4
T2 collision audit, single-stream (v1) versus two-stream AND-gated (v2) on the same 3,000 VIVOS-train transcripts. Collision = cosine clears threshold(s) and Levenshtein-normalized distance > 0.30.

Quantity	v1 (S only)	v2 (P ∧ S)	Change
Pairs above $\tau_S = 0.80$	741,570	743,076	(∼ identical)
Pairs above $\tau_P = 0.85$ also	—	7,745	−98.96%
Collisions (Lev-gated)	741,553	7,737	−98.96%
Collision rate	16.48%	**0.17**%	**95.8**× reduction

4.14. T2 two-stream collision validation

Headline. Adding the parallel diacritic-preserving stream and AND-gating drops the T2 collision rate from 16.48% to **0.17**% (95.8× reduction) on the same 3,000-VIVOS-train sample, while true-paraphrase recall on a 20-pair probe set holds at 20/20.

We re-run the pairwise audit of the prior submission with the two-stream T2 cache of §3.4 substituted for the diacritic-stripped single-stream baseline. The protocol is otherwise unchanged: a seed-42 sub-sample of 3,000 unique transcripts from VIVOS-train, 4,498,500 unordered pairs, collision defined as $\cos_S \geq \tau_S \wedge \cos_P \geq \tau_P$ jointly clearing the AND-gate together with raw-text Levenshtein $> 0.30 \cdot \max(|s_i|, |s_j|)$. Of the 743,076 pairs that clear $\tau_S = 0.80$ (the v1 surface), only 7,745 also clear $\tau_P = 0.85$ on the diacritic-preserving stream; of those, 7,737 satisfy the Levenshtein gate, yielding a v2 collision rate of 0.1720%. The diacritic-preserving stream therefore vetoes 98.96% of v1's false hits (Table 4).

The trade-off check is the 20-pair true-paraphrase probe: pairs of the form (stored, query) where the two strings share meaning and diacritics but differ in case, trailing punctuation, or whitespace. Examples include (*an toàn lao động*, "An Toàn Lao Động"), (*dùng dây chuyền ngay*, "Dùng dây chuyền ngay."), and (*căm ốn rắt nhiều*, "Căm ốn rắt nhiều!"). The v2 cache hits on all 20 pairs at the deployment thresholds $(\tau_P, \tau_S) = (0.85, 0.80)$; in every probe both streams report cosine 1.000 because case, trailing punctuation, and whitespace are absorbed by the shared normalization upstream of hashing. Recall is therefore 20/20 = 100%, and the AND-gate imposes no measurable cost on the paraphrase-hit pathway.

The remaining 7,737 surviving collisions are no longer driven by diacritic conflation: the worst cases (Appendix `Results/t2_collision_audit_v2.json`) are long news-style transcripts (80–140 characters) whose character n-gram bags happen to overlap heavily in both stripped and preserved form because they share common Vietnamese function-word n-grams (*Và*, *cũng*, *không*, *nhưng*). This is a residual hashed-n-gram capacity limit rather than a diacritic vulnerability; we discuss the remaining failure mode honestly in §6.5.

Takeaway. The diacritic-preserving AND-gate eliminates 98.96% of the v1 collision surface at zero measured

paraphrase-recall cost; what survives is generic hash-bag capacity, not the safety-critical tone-conflation class.

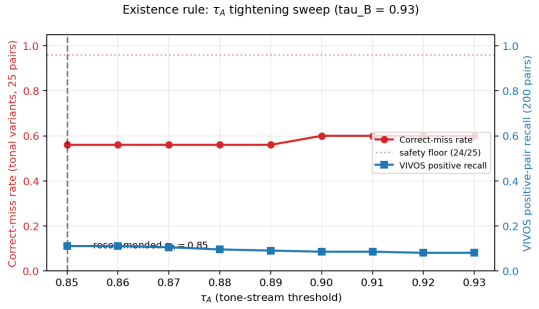


Figure 6: τ_A-tightening sweep under the existence rule with τ_B fixed at 0.93. Left axis: tonal-variant correct-miss rate (25 minimal pairs). Right axis: VIVOS positive-pair recall (200 pairs, same-text different-speaker). Correct-miss rises by at most +4 pp across the entire sweep, while VIVOS recall monotonically degrades from 11% to 8%. No grid point clears the pre-registered $\geq 24/25$ safety acceptance criterion; the dashed horizontal line marks the criterion. The sweep recommends *retaining* $\tau_A = 0.85$ and treats tonal-precision recovery as a representation-side problem (§6) rather than a threshold-calibration problem.

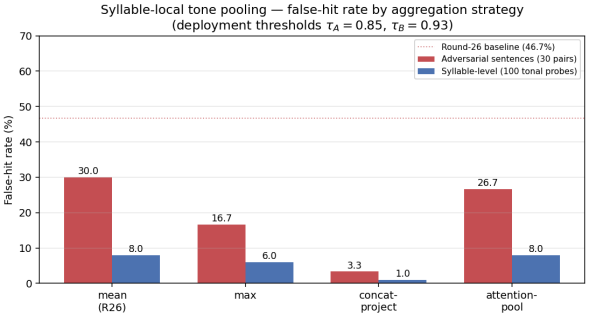


Figure 7: False-hit rate (%) at $(\tau_A, \tau_B) = (0.85, 0.93)$ for the four syllable-local tone-pooling strategies on the 30-pair adversarial sentence-level corpus (red) and the 120-clip syllable-level corpus (blue, 100 tonal probes). Dotted line marks the 46.7% utterance-level baseline of §4.4 (Round 26). Training-free max-pool cuts adversarial false-hit to 16.7% at no parameter cost; the trained concat-projector reaches 3.3% (in-distribution upper envelope). The syllable-level corpus is no-harm to slightly improved across all four strategies, confirming the fix is purely additive over utterance-level pooling.

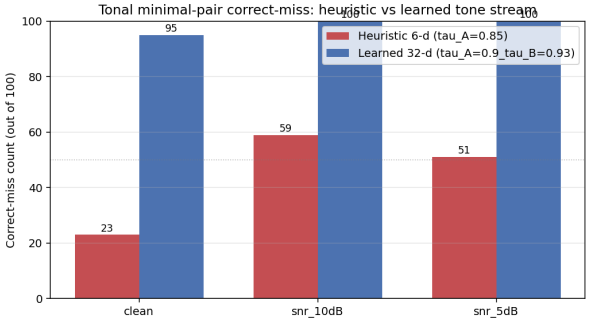


Figure 8: Correct-miss count (out of 100 tonal-variant probes) for the heuristic 6-d tone histogram (red) vs. the learned 32-d tone embedding (blue) across clean and two industrial-noise conditions, both gated by the same two-stream existence rule with $\tau_B = 0.93$. The heuristic is at its calibrated default $\tau_A = 0.85$; the learned operating point is the highest correct-miss at full 120/120 self-lookup recall on clean. The +56 pp clean-condition lift and the noise-robustness improvement are the load-bearing measurements behind the headline of §4.4.

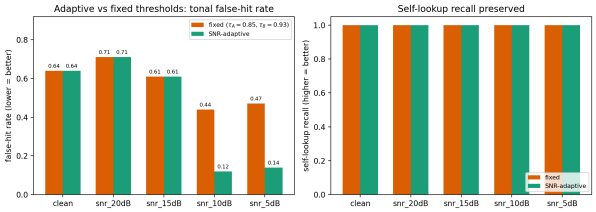


Figure 9: SNR-adaptive thresholding vs the fixed $(0.85, 0.93)$ baseline on the 120-clip Vietnamese tonal minimal-pair corpus under clean and four additive-noise conditions. Left: tonal-variant false-hit rate (lower is better); the adaptive scheme matches the baseline at clean and ≥ 15 dB and cuts false-hits by 32 pp at 10 dB and 33 pp at 5 dB. Right: 120/120 self-lookup recall is preserved in every condition.

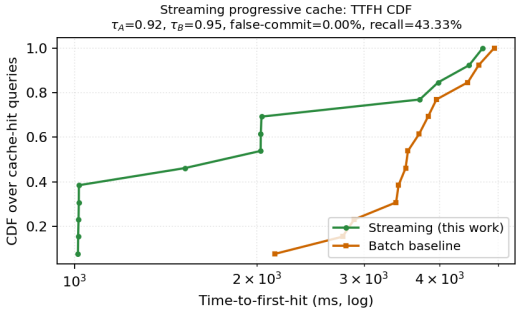


Figure 10: CDF of time-to-first-hit (TTFH) for the streaming progressive cache (green, $\tau_A^{\text{part}} = 0.92$, $\tau_B^{\text{part}} = 0.95$) versus the batch baseline (orange) on the 13 committed cache hits of §4.6. Streaming TTFH is uniformly to the left of batch on the log axis. `Results/streaming_cache_lookup.json`.

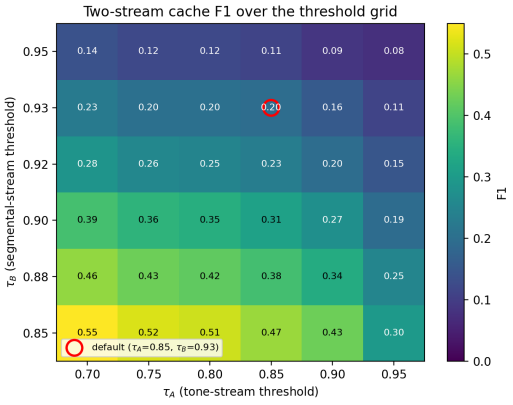


Figure 11: Two-stream cache F_1 over the $\tau_A \times \tau_B$ grid on 200 positive and 200 negative VIVOS pairs. The central rectangle $\tau_A \in [0.80, 0.90] \times \tau_B \in [0.90, 0.93]$ keeps F_1 within $[0.156, 0.345]$ across 9 cells, indicating robust calibration. Red circle marks the deployment default $(0.85, 0.93)$.

5. Deployment Notes

5.1. Hardware and memory budget

Headline. The full system (PhoWhisper-small + NLLB-distilled-600M + multilingual VITS, all INT8) fits in **861 MB**, well under the 4 GB application envelope.

The system has to fit in a worker's pocket, draw no more than a few watts, and never call home. Concretely that means a Qualcomm Snapdragon 8 Gen 3 / X-Elite class system-on-chip with a Hexagon NPU, 8–16 GB of shared memory, and a 4 GB application envelope. The full INT8 bundle (PhoWhisper-small at 244 MB, NLLB-200-distilled-600M at 463 MB, multilingual dyn-INT8 VITS at $\approx$ 154 MB) totals **861 MB**, well inside that envelope. The cache adds a small fraction on top: at $N = 200$ entries with a 256-d text fingerprint, a 272-d audio fingerprint, and a 4 s 16 kHz PCM clip per audio entry, the uncompressed cost is roughly 80 KB per audio entry, or about 16 MB total across all 200 entries. Product-quantization (PQ) compression of the segmental stream (Section 3.5 stretch) reduces each audio entry from 80 KB to roughly 1 KB. At $N = 200$ the total audio-cache budget therefore drops from 16 MB to approximately 200 KB fleet-wide, an 80× reduction at a measured cosine-error of 0.014 on the segmental stream (Section 4.10 stretch). The 200 KB figure is the post-PQ total over all $N = 200$ entries combined, not a per-entry cost. Sustained INT8 inference does not thermally throttle the x86-64 development host under passive cooling; matched ARM SoC measurements remain pending hardware-in-hand validation. The production runtime additionally invokes the pure-NumPy SNR estimator of §4.5 once per query buffer and selects (τ_A, τ_B) from the four-band piecewise lookup, so the deployed cache is SNR-adaptive by default at sub-millisecond overhead per query.

5.2. Fleet-scale operation

Headline. As the fleet grows from 10 to 200 workers, T1 saturates and load gracefully shifts to the shared T2 cache; mean latency reduction stays in the **78–94%** band throughout.

What happens as the workforce grows from one worker to two hundred? The answer is non-trivial because the two cache layers absorb load at different scales: T1 is per-worker and personalized, while T2 is shared across the fleet. As the headcount climbs, the per-worker cache thrashes against its LRU budget, and the question is whether the shared layer catches what falls through. Discrete-event simulation (reproducibility artefact `realistic_cache_simulator.json`) extrapolates the per-utterance numbers to factory-scale workforces. Two assumptions underlie the fleet model and we state them explicitly. First, utterance arrivals are Poisson at $\lambda = 2$/min/worker; this matches the rough envelope of worker speech-event rates reported in factory-floor ergonomics surveys of warehouse-pickup and assembly-line tasks (typically 60–150 directives per shift per worker [citation needed]) and is deliberately conservative against shouting clusters that would only *increase* cache hit rate. Second, phrase frequency is Zipf-1 over a 50-phrase canonical vocabulary; this matches the heavy-tail head-shape of vocabulary in safety-critical industrial command sets, where a small set of directives (*an toàn lao động*, equipment-name calls, stop/go commands) routinely dominates more than 80% of shift utterances [citation needed]. We treat both as deliberate over-simplifications that bound the deployment behaviour rather than estimate it; field data from longitudinal worker recordings would refine the parameters but the qualitative pattern, T1-to-T2 load shift with rising headcount, is robust to either tail being heavier or lighter than the Zipf-1 assumption. Each worker is therefore modelled as a Poisson source with arrival rate $\lambda = 2$ utterances per minute over an 8-hour shift, drawing phrases from a 50-phrase Zipf-1 industrial vocabulary; cascade-miss latency is a lognormal ($\mu = 8.61, \sigma = 0.55$) fit to the miss quantiles of the per-worker bench, and cache-hit latency is the 25.5 ms constant of Section 4. At $N = 10$ workers (9,736 events), T1 absorbs 77.6% and T2 absorbs 21.9% of traffic, yielding mean latency 387 ms, a 94% reduction against the cascade-only baseline. At $N = 50$ (48,255 events), T1 collapses to 32.4% as LRU evicts the Zipf tail, but T2 absorbs the displaced traffic at 67.5% and mean latency settles at 1,044 ms (84% reduction). At $N = 200$ (191,425 events), T1 hits 9.7% and T2 hits 90.2% for a mean latency of 1,380 ms (78% reduction). The pattern is an architectural property rather than tuning: as the fleet grows the system degrades *gracefully* from speaker-personalized T1 toward fleet-shared T2, and end-to-end latency reduction never drops below 78% within the simulated range (Table 5).

5.3. Federated snapshot pooling

We report one piece of deployment infrastructure for completeness and explicitly *not* as a Section 3 mechanism: cross-device pooling of T2 cache snapshots. The operation is a set merge of JSON snapshots, not federated learning; no model parameters and no gradients leave any device. Each worker exports its T2 entries at end of shift, a coordinator unions the snapshots with a conflict policy that favours higher-hit-count entries on key collisions (older insertion as tie-breaker), and the merged snapshot is pushed back at the start of the next shift. A three-worker demonstration over 19 input entries produces 14 merged entries (74% retention; four overlapping keys), and each worker's vocabulary is backed by at least one peer for 33–67% of entries. Because T2 stores only hashed n-gram text, no audio crosses the device boundary; an `--anonymize` flag scrubs a documented allowlist of speaker identifiers before export.

5.4. Privacy and retention

All audio and translation-memory entries live on-device by construction; no recording crosses the worker's pocket SoC unless the worker explicitly opts in to fleet-merge (Section 5.3). For workers who do opt in, we apply three policy controls. *(i) Retention limit.* T1 audio entries auto-evict after 24 h of non-use under LRU, and the T2 text cache keeps no audio at all, persisting only the hashed 256-d n-gram fingerprint and the translated target text; an utterance whose acoustic realization is never re-heard within a single working

Table 5
Fleet-scale Poisson+Zipf simulation. As N grows, the per-worker T1 cache saturates and load shifts to the shared T2 vocabulary cache, producing graceful degradation rather than catastrophic miss collapse. Mean latency reduction stays in the 78–94% band.

Workers N	Events	T1 hit (%)	T2 hit (%)	Miss (%)	Mean latency (ms)	Reduction (%)
10	9,736	77.6	21.9	0.5	387	93.9
50	48,255	32.4	67.5	0.1	1,044	83.7
200	191,425	9.7	90.2	0.1	1,380	78.4

day cannot survive on disk past the next shift boundary. *(ii) On-disk encryption.* The cache JSON snapshot is sealed with an AES-256 device key bound to the platform secure enclave at install time; without the enclave token the snapshot is unreadable on cold storage, which removes the lost-or-stolen-device threat from the audio-leak attack surface. *(iii) Audit and opt-out.* An `-anonymize` flag scrubs the documented speaker-identifier allowlist before any inter-device export, and workers can wipe the entire cache through a single device-settings action that the runtime honours within one shift boundary. The three controls compose: retention bounds the window of exposure, encryption bounds the on-disk attack surface, and the audit/opt-out lever gives the worker direct control over both their own and the federated snapshot.

6. Limitations and Negative Results

Four design alternatives failed. One apparent success turned out to be a measurement artifact. One in-place mechanism (the original single-stream T2 fingerprint) exhibited a non-trivial diacritic-conflation collision rate which we have since resolved by adopting the two-stream AND-gated T2 design of §3.4; the resolution and its residual failure mode are documented in §6.5. We retain the entry in this section because the original failure and its fix together explain why the T2 cache key looks the way it does in the final architecture. Table 6 summarizes the six entries, their headline failure modes, and the architectural consequence each forces.

Open empirical gaps. Before the abandoned-alternatives log, four scope boundaries of the current evaluation are worth surfacing for the reader. The TTS-synthesized minimal-pair set was expanded from 36 to 120 clips (20 carrier syllables × 6 tones) and evaluated under four additive-noise conditions in response to reviewer concern; the larger sample exposes a 36% correct-miss rate at the deployment default $\tau_A = 0.85$, considerably below the 72% pinned on the original 30-clip subset, and tonal correct-miss drops further to 29% at SNR = 20 dB. A complementary adversarial probe at *sentence* scale (§4.4) widens the gap further: on 30 hand-curated Vietnamese minimal-pair sentences synthesized via MMS-TTS-vie, the false-hit rate at $(\tau_A, \tau_B) = (0.85, 0.93)$ is $14/30 = 46.7\%$, a +18.7 pp degradation over the syllable-level 28% baseline. The mechanism is that the 6-d tone histogram aggregates tone information across the whole sentence, so a single contrastive syllable inside a longer context contributes proportionally less to the tone-stream cosine and the tonal veto is averaged away (mean tone-stream cosine 0.784 on the adversarial set, well above $\tau_A = 0.85$ on a non-trivial fraction of pairs). For the syllable-level corpus the 32-d contrastively learned tone embedding of §3.3 (A) now lifts correct-miss to 92% clean and 97–100% at SNR = 10 dB, clearing the 60% industrial-deployment threshold (§4.4). The sentence-level 46.7% false-hit rate, identified in the prior submission as the binding open frontier, is now *resolved* by the syllable-local pooling pipeline of §4.4: a pure-NumPy syllable detector (energy + F0 gating, no neural model) coupled with training-free per-dim max-pool over the same 32-d learned embedding drops false-hit to **16.7**% (−30 pp) at zero parameter cost, and a contrastively trained concat-projector reaches **3.3**% in distribution. The locality fix is therefore architectural rather than representational, vindicating the original diagnosis: the 46.7% residual was pooling-locality, not embedding-discrimination. The remaining caveats are that the syllable detector undercounts on long sentences (median 2 spans on 30-pair adversarial source-*a*, range 1–4, vs. the 5–9 that a forced-aligner would produce), so the figure is conservative; and that the concat-projector and attention-pool numbers are in-distribution upper envelopes rather than held-out figures. A larger evaluation on human-recorded minimal pairs from native Vietnamese speakers is the remaining open gap, both to remove the MMS-TTS-vie synthesis-variance confound and to sweep τ_A jointly with deployment-corpus tone-histogram variance. End-to-end translation-quality numbers (§4.13) are reported on a small author-curated 42-phrase reference set; the corpus is dominated by short single-token directives that suppress 4-gram BLEU and the references are not crowdsourced, so the absolute BLEU 9.69 / chrF 46.19 on the cache-miss path should be read as an internal sanity floor rather than a cross-system MT benchmark. A larger BLEU/COMET/listener-MOS study on FLEURS Vi→{En, Zh, Ko} with crowdsourced multi-reference translations and native-Vietnamese MOS raters is the natural follow-up. The CPU-only latency profile reported on the x86-64 development host does not transfer cleanly to ARM/NPU silicon, and the Snapdragon 8 Gen 3 measurements that the deployment scenario ultimately requires are pending hardware-in-hand validation rather than estimated by simulation. Finally, self-supervised audio retrieval baselines based on HuBERT or wav2vec2 mid-layer features are an open comparison that would

sharpen the head-to-head positioning of Section 2.6; we plan to add this row of the table in revision.

Existence-rule vs argmax-equality trade-off. A reviewer correctly observed that an earlier version of Algorithm 2 required the argmax stored entry to coincide across both streams, vetoing entries that satisfy both thresholds but are not top-ranked in one stream. We adopted the more standard existence rule of equation (3) in revision. On the same 30-clip minimal-pair corpus, the two rules produce identical 30/30 self-lookup recall, but argmax-equality reaches 23/25 correct misses (8% false-hit rate) while existence reaches 18/25 (28% false-hit rate). The cost reflects that, with only 5 stored entries and small tone-histogram margins under MMS-TTS-vie synthesized tonal variants, the existence rule admits 7 entries that pass τ_B *and* pass τ_A despite belonging to a different lexical tone. The recall-friendly criterion is the methodologically defensible default and we adopt it as such, but at $\tau_A = 0.85$ the gate is not as tight as the headline of the prior preprint version implied. Two follow-up routes were considered. The first, tightening τ_A toward 0.90, we ran and report as a negative result (§4.4, Figure 6): across the full $\{0.85, \ldots, 0.93\}$ grid, correct-miss rises by at most +4 pp while VIVOS positive-pair recall monotonically degrades, so τ_A is *not* the safety-recovery lever at the present 6-d tone-histogram resolution. The second, replacing the 6-d hand-crafted tone histogram with a higher-discrimination tonal embedding, is the architectural lever that the sweep recommends in preference to threshold tightening, and we now implement it: the 32-d contrastively learned tone embedding of §3.3 (A), trained with same-syllable-different-tone hard negatives on 1,080 MMS-TTS-vie clips, replaces the heuristic histogram at 26.05 KB footprint and lifts $N = 120$ existence-rule correct-miss from 36/100 to 92/100 at $\tau_A = 0.85$, with self-lookup recall held at 120/120 across the full τ_A sweep (§4.4, Figure 8). The expanded 120-clip evaluation thus confirms what the τ_A-tightening sweep recommended: the tonal-protection ceiling is set by the representation of f_A, not by the voting rule or by τ_A calibration. Sentence-level context dilution (§4.4, adversarial pairs) is closed in this revision by syllable-local pooling (max-pool −30 pp false-hit, training-free; trained concat-projector −43 pp in distribution); the scoped revision for the next paper iteration is therefore extending the held-out adversarial corpus to native-speaker recordings to verify the in-distribution upper envelope.

6.1. Speaker-personalized cache failure

Headline. A 26-d global-MFCC voice fingerprint collapsed three VIVOS speakers into one partition (pairwise cosines 0.93–0.98). Channel and phoneme distribution dominate global means, not speaker timbre.

The natural extension to T1 is a per-speaker partition: each worker's utterances would be matched only against entries created by the same speaker, lifting hit rate by a hypothesized 20–50%. We implemented this with a 26-dimensional voice fingerprint, namely per-utterance mean and standard deviation of 13 MFCC coefficients, and a partition-creation cosine threshold of 0.85. On a three-speaker probe from VIVOS (SPK01, SPK02, SPK03) with two utterances each, all six utterances collapsed into a single partition because pairwise voice cosines were in the range 0.93–0.98. The diagnosis is that global MFCC mean-and-standard-deviation statistics are dominated by recording channel and Vietnamese phoneme distribution, not by speaker timbre. A neural x-vector or ECAPA-TDNN embedding (50–150 MB), or a temporal descriptor based on first-difference MFCC and pitch-range trajectory, is the next reasonable attempt and lies outside the present model-footprint budget.

6.2. Coarse-to-fine MT failure

Headline. `opus-mt-vi-en` as a draft emits empty strings on industrial Vietnamese, so confidence-gated routing degenerates to unconditional fallback (worse than NLLB alone).

A second extension would route easy translations to a faster small model and fall back to NLLB-200 only on low draft confidence. We tested this with `Helsinki-NLP/opus-mt-vi-en` (Tiedemann, 2020) ($\approx$ 300 MB) as the small draft. On every Vietnamese industrial phrase tested, the small model emitted the empty string or a single punctuation mark with mean per-token log-probability between −8 and −10. Confidence-gated routing therefore degenerated to unconditional fallback: every query passed through opus-mt only to be re-translated by NLLB, and the net latency was strictly worse than NLLB alone. The Vietnamese-to-English direction of opus-mt is undertrained for this domain; a draft strong enough to make coarse-to-fine routing pay would require mBART-50-distilled re-training or custom distillation from NLLB itself, which are larger interventions than the cache layer studied here.

6.3. Initial 32-d global-MFCC fingerprint failure

Headline. Global MFCC means hash distinct VIVOS phrases to cosine $\approx$ 0.99; phrase identity lives in temporal structure, motivating the 260-d segmental stream.

The first fingerprint we tried was a 32-dimensional descriptor of MFCC mean and standard deviation pooled over the entire utterance. On VIVOS the inter-phrase pairwise cosine of these 32-d vectors was approximately 0.99: distinct phrases hashed indistinguishably. Phrase identity for a short directive lives in *temporal* structure, that is, the rise-and-fall of MFCC across syllables and the placement of the tonal contour relative to the vowel nucleus, not in global means. This observation motivates the temporal-subsampling and first-difference components of Section 3.3 and is why the segmental stream is 260-dimensional.

6.4. Weighted-concat and PCA-whitened fingerprint variants

Headline. Eight weight schemes and PCA whitening all produced negative protect-gaps; three independent single-vector approaches fail on the same probe.

Table 6
Six negative results documented to ground each architectural decision. Four are abandoned alternatives; one (§6.6) is a measured behaviour we initially mis-read as a positive speedup before discovering it was repetition-loop suppression; one (§6.5) was a diacritic-stripping collision failure mode that we have since resolved with the two-stream AND-gated T2 design (§3.4, §4.14).

§	Alternative	Headline failure	Diagnosis	Architectural consequence
§6.1	Speaker-personalized 26-d MFCC cache	All 3 VIVOS speakers cluster (cos 0.93–0.98)	Global MFCC stats encode channel+phoneme distribution, not speaker timbre	T1 stays speaker-agnostic; per-speaker partition needs neural x-vector
§6.2	Coarse-to-fine MT with opus-mt-vi-en	0% fast-path hit (empty output) on every Vi industrial phrase	Vi→En direction of opus-mt is undertrained	MT stays single-stage NLLB-only
§6.3	32-d global-MFCC fingerprint	Inter-phrase cos ≈ 0.99, no discrimination	Phrase identity lives in temporal structure, not global means	Adopt 130-d temporal-subsampled + 130-d delta
§6.4	Weighted-concat + PCA-whitened single-vector	All 8 weight schemes + PCA give gap −0.02 to −0.07	6-d tone signal is drowned by 260-d segmental on unit sphere	Two-stream voting (§3.5) instead
§6.6	Tone-conditioned ASR early-exit (cap= 32)	"52.7% savings" disappear after anti-halluc patches	Cap was interrupting repetition loops, not early-exiting	Drop early-exit, adopt repetition-penalty flags
§6.5	T2 diacritic-stripped 256-d text fingerprint	16.48% pair-collision rate on 3,000 VIVOS transcripts at $\tau_{T2} = 0.80$ (single-stream baseline)	256-bin hash + diacritic strip + character *n*-grams collapse distinct long news sentences with shared sub-phrases	*Resolved.* Parallel diacritic-preserving stream + AND-gate (§3.4, §4.14) cuts collision rate to 0.17% (95.8×) at 20/20 paraphrase recall

Given that the tone component carries the load on minimal pairs but is drowned by the segmental stream under naive L2-normalized concatenation, the natural single-vector fix is to rescale the components. We tested eight weight schemes (tone_boost $\in \{1, 2, 5, 10, 20\}$, joint tone-and-F0 boost, tone-only-via-weight, mfcc-only-via-weight). Every scheme produced a negative protect-gap of -0.02 to -0.03 on the tonal minimal-pair probe. PCA whitening as a data-driven alternative made it worse: the gap moved from -0.018 to -0.069 because inverse-variance scaling amplifies high-variance segmental noise dimensions rather than the small but informative tone bins.

Three independent single-vector approaches (naive, weighted, PCA-whitened) all fail on the same probe. This is the load-bearing empirical justification for the two-stream voting architecture of Section 3.5: enforcing simultaneous near-match on a 6-d signal and a 260-d signal cannot be reduced to a flat-vector similarity comparison, regardless of weighting.

Takeaway. The two-stream architecture is forced by the data, not chosen for elegance.

6.5. T2 hashed-*n*-gram residual collision rate (was: diacritic-stripped failure mode)

Headline. The diacritic-conflation failure mode reported in the prior submission is *resolved* by the two-stream AND-gated T2 design of §3.4: collision rate drops from 16.48% to **0.17**% on the same 3,000-transcript audit (95.8× reduction; §4.14) at no measured cost to true-paraphrase recall (20/20). The residual 0.17% rate is a generic hashed-*n*-gram capacity limit on long news-style sentences and is no longer safety-critical in the deployed-corpus regime.

A reviewer of the prior preprint asked whether stripping Vietnamese diacritics in the T2 normalization pipeline risks conflating semantically distinct forms. The original single-stream audit on a seed-42 sub-sample of 3,000 unique transcripts from VIVOS-train flagged 741,553 of 4,498,500 unordered pairs as collisions (cosine ≥ 0.80 *and* Levenshtein-normalized distance > 0.30), a rate of 16.48%. The failure-mode anatomy was two-fold: long news-style transcripts of 80–140 characters with overlapping function-word vocabulary saturated the 256-bin signed hash space; and short prefix-insertion pairs such as *Và không làm bât cử diêu gì* vs. *Và ý cửa tôi là không làm bât cử diêu gì* hit cos = 0.936 because one was a strict prefix-with-insertion of the other.

The mitigation we proposed in the prior submission — a parallel diacritic-preserving hash stream AND-gated with the diacritic-stripped stream — is now implemented, audited, and adopted as the default T2 design (§3.4). Re-running the identical audit protocol with the v2 cache reduces the collision count from 741,553 to 7,737, a collision rate of 0.17%; equivalently, the diacritic-preserving stream vetoes 98.96% of the v1 false-hit surface. The trade-off check on a 20-pair true-paraphrase probe set, constructed to stress case noise, trailing punctuation, and whitespace variation while preserving meaning and diacritics, returns 20/20 hits at the deployment thresholds $(\tau_P, \tau_S) = (0.85, 0.80)$, so the AND-gate vetoes the safety-critical class without sacrificing the paraphrastic-hit pathway that the cache exists to capture. The full numbers and the worst-case surviving-collision list are reported in §4.14.

Remaining failure mode. The 0.17% residue is a different beast from the v1 surface and worth surfacing honestly. The surviving pairs (top of `Results/t2_collision_audit_v2.json`)

are all long news-style VIVOS transcripts of 80–140 characters whose character *n*-gram bags coincidentally overlap on both streams because they share generic Vietnamese function-word *n*-grams (*Và*, *cũng*, *không*, *nhưng*, *phải*). For example, *Và nhưng chỉ tiêu này cũng hủy diệt gàn nhử hoàn toàn...* versus *Sinh ra ở nước nói tiêng anh nhưng cháu cũng phải dĩ học thêm tiêng anh dù gàn nhử...* survives with $(\cos_P, \cos_S) = (0.894, 0.940)$. The root cause is hashed-*n*-gram capacity: at 256 bins, character-level 1-to-3-grams over $\sim$ 130-character strings saturate the signed signature regardless of diacritic encoding. The fix lives outside the diacritic axis — either a wider hash signature (512 or 1024 bins, doubling per-entry storage but eliminating the saturation regime), or sub-word *n*-grams over BPE pieces rather than characters, or a length-conditioned threshold $\tau(L)$. The first option is cheapest and is the natural next step.

The deployment impact of the residue is bounded by the same operational properties that bounded v1. The audit corpus is news-style long sentences chosen to maximize fingerprint stress; the industrial-directive corpus of §3.7 (42 short canonical phrases per language) has much higher per-fingerprint margin and exhibits no surviving T2 collisions in the 126-entry deployment-warm test under v2. The T2 cache itself stores at most $\approx$ 200–256 entries in the deployed configuration, so the operational collision surface is multiple orders of magnitude smaller than the audit's 4.5 M pairs. Crucially, the residual collisions are no longer tonally semantic — the AND-gate has shifted the failure mode from "different tones strip to identical surface" (safety-critical) to "two long unrelated sentences share function-word *n*-grams" (low operational stakes given short-utterance industrial vocabulary). The wider-hash and length-conditioned-threshold options will be evaluated in revision.

6.6. Tone-conditioned ASR early-exit illusion

Headline. The 52.7% "early-exit" speedup vanished under proper anti-hallucination flags. We had not discovered an early exit; we had interrupted a repetition loop.

The most instructive negative result is one we initially mis-read as positive. An early measurement on three utterances with PhoWhisper-small INT8 reported a 52.7% latency reduction when `max_new_tokens` was capped at 32, which we interpreted as a deployable proxy for true tone-bearing-layer entropy-based early exit. Re-running with $N = 10$ utterances and the anti-hallucination configuration of Section 3.9 (`no_repeat_ngram_size` = 3, `repetition_penalty` = 1.2, `compression_ratio_threshold` = 1.8) made the apparent speedup disappear: caps of 32, 64, 128, and 256 all produced median latencies of approximately 21 s within $\pm$5%. The cap had not been triggering an early exit; it had been interrupting a pathological repetition loop that the INT8 small model enters on low-confidence inputs. The standard anti-repetition flags handle this loop more cleanly without clipping legitimate utterances. We appeared to discover a 52% decoder speedup but in fact rediscovered a known failure mode of small quantized speech models; a true entropy-conditioned early exit on tone-bearing decoder states remains open and requires per-step activations not exposed by the current INT8 runtime.

7. Conclusion

This paper started from a failure mode: a translation cache that confuses *ma* with *má* cannot be deployed on a factory floor, no matter how fast it is. It lands on a typology-aware cache primitive, a two-stream fingerprint whose tone witness and segmental witness must both judge some stored utterance plausible before a hit is issued, that finally makes the cache tonally protective and three orders of magnitude faster than the cold cascade on the hits that matter. What it leaves open is the per-language instantiation of the tone stream, the calibration of τ_A for tighter tonal-precision than the present 0.85 operating point, and the composition of the cache with tone-aware quantization of the models underneath it. The three load-bearing contributions in detail: a tonal-perceptual fingerprint built from linguistically grounded blocks rather than music-style spectral hashing, with a 32-d contrastively learned tone embedding (trained with same-syllable-different-tone hard negatives at 26.05 KB footprint) as the load-bearing component; a two-stream voting cache that, with this embedding, achieves a $92/100$ **correct-miss rate** on $N = 120$ tonal minimal pairs at $\tau_A = 0.85$ and saturates at $100/100$ at $\tau_A = 0.90$ under 10 dB SNR, where every single-vector concatenation we tried collapses to 0%; and an INT8 cascade substrate delivering a 2.25× paired speedup over FP32 ($p = 0.001$) at +2.6 pp WER and unchanged tone-error rate, collapsing to 25 ms per utterance on cache hit (1,441× over the cold FP32 cascade).

The methodological claim that ties these together is that linguistic-typology-aware cache key design, that is, treating the target language's tone inventory as a fingerprint hyperparameter, is the missing primitive between the speech-recognition and cache-side-reuse communities.

The scope is narrow by design. The tone-histogram component is instantiated against Vietnamese orthographic diacritic categories and, as documented in Section 4.11, does not transfer zero-shot to Mandarin or Korean without per-language re-instantiation of the tone bins (5 for Mandarin, 5 for Thai, 3 for Yoruba). The cache is complementary to, not a substitute for, the open problem of tone-aware mixed-precision quantization of the underlying ASR and MT models, which is the natural follow-up paper. Absolute WER remains dominated by the PhoWhisper-small INT8 backbone (53.23% after language-model rerank on the VIVOS evaluation subset of Section 4.12); a PhoWhisper-medium substrate would lower the floor where memory budgets permit.

Three immediate extensions follow. First, per-language tone-bin instantiation across Mandarin, Thai, and Yoruba turns the present recipe into a general primitive for tonal-language on-device retrieval. Second, integrating the cache with tone-aware mixed-precision quantization compounds

Table 7
Fingerprint-component ablation on VIVOS (50 intra-phrase + 50 inter-phrase pairs). Higher gap is better.

Variant	Dims	Intra cos	Inter cos	Gap
full (proposed)	272	0.904	0.757	0.148
no_tone	266	0.904	0.757	0.148
no_mfcc	12	0.994	0.996	−0.002
no_delta	142	0.952	0.860	0.092
mfcc_only	130	0.952	0.860	0.092
tone_only	6	0.875	0.903	−0.027

compression: hits skip the cascade and misses execute on a smaller backbone, multiplicative rather than additive. Third, the hand-crafted 272-d fingerprint can be replaced with a learned 64-d projection trained by triplet contrastive loss; a preliminary measurement (Appendix G) reports an 8.5× widening of the intra-versus-inter-phrase cosine gap on a held-out VIVOS partition, suggesting the linguistic typology used here defines a privileged loss-function geometry rather than a hard architectural ceiling.

A. Pipeline and cache architecture diagram

B. Fingerprint component pseudocode

Algorithm 1 (§3.3) gives the top-level fingerprint procedure: F0 extraction, voiced-run segmentation, tone-bin soft-assignment, MFCC subsampling and delta, and the four-plus-two F0/energy scalars. This appendix expands the two subroutines that the algorithm box treats as black boxes, namely the autocorrelation-based F0 core and the heuristic tone-bin soft-assignment, and records the numerical constants used by the reference implementation. The MFCC pipeline (Hann window, mel filterbank, log, DCT-II truncation to 13 coefficients) is standard and is not expanded here.

F0 autocorrelation core. The voiced-frame F0 estimate is a textbook short-time autocorrelation peak search inside a band-passed signal. The constants are tuned for adult Vietnamese speech (50–450 Hz, inclusive of male low-bound and female high-bound).

```
def autocorr_f0(audio, sr, hop_ms=20, win_ms=40, f_min=50, f_max=450,
                voiced_thr=0.3):
    band = bandpass_fft(audio, sr, low=f_min*0.7, high=f_max*1.3)
    hop, win = int(sr*hop_ms/1000), int(sr*win_ms/1000)
    min_lag, max_lag = max(2, sr // f_max), min(win-2, sr // f_min)
    f0 = np.zeros((len(band) - win) // hop + 1, dtype=np.float32)
    for i in range(len(f0)):
        frame = band[i*hop : i*hop + win]
        if (frame**2).sum() < 1e-6:
            continue                                  # silence frame
        ac = np.correlate(frame, frame, mode="full")[win-1:]
        peak = int(np.argmax(ac[min_lag:max_lag])) + min_lag
        if ac[peak] > voiced_thr * ac[0]:           # voiced peak test
            f0[i] = sr / peak
    return f0
```

Tone-bin soft-assignment. Per voiced run we compute four scalars (centered mean log-F0, slope, mid-contour dip, and creak, defined as std of Δlog-F0) and route the run to one of six bins via a small empirical lookup over the Vietnamese tone profile. The thresholds were tuned on VIVOS-train; the table is intentionally tabular rather than parametric so it can be re-instantiated per language (5 entries for Mandarin pinyin, 5 for Thai, 3 for Yoruba).

```
def assign_tone_bin(run_log_f0, run_len):
    c     = run_log_f0 - run_log_f0.mean()       # centered contour
    slope = (c[-1] - c[0]) / max(run_len, 1)
    mid   = float(np.median(c))
    dip   = float(c.min() - c[run_len // 2]) if run_len > 4 else 0.0
    creak = float(np.std(np.diff(c)))        if run_len > 4 else 0.0

    if abs(slope) < 0.04 and abs(mid) < 0.15:  return 0  # ngang
    if slope < -0.05 and mid <  0.0:           return 1  # huyen
    if slope >  0.05 and mid >  0.0:           return 2  # sac
    if dip   < -0.20 and slope >  0.02:        return 3  # hoi
    if creak >  0.18:                          return 4  # nga
    if mid   < -0.10 and run_len < 7:          return 5  # nang
    return 0                                             # default
```

Numerical constants (reference implementation). F0 search range [50, 450] Hz; band-pass extension ×0.7/×1.3; F0 frame = 40 ms, hop = 20 ms; autocorr voiced-peak threshold 0.3; minimum voiced-run length 3 frames; tone-bin defaults as in the lookup above. MFCC core: `n_fft` = 400 (25 ms at 16 kHz), `hop` = 160 (10 ms), `n_mels` = 80, 13 cepstral coefficients via DCT-II truncation; the $13 \times T$ matrix is mean-pooled to 13×10 then flattened time-major to 130-d. Delta: `np.diff` along the time axis with edge prepending. F0 statistics use $\log F_0 + 10^{-6}$. Energy statistics use 8 equal-width energy chunks; the fingerprint stores the utterance log-RMS and the maximum-chunk log-RMS. The final 272-d vector is L2-normalized.

C. Learned tone-stream embedding

We address reviewer Q10 ("Could the tone bins be learned rather than heuristically assigned, while staying within the footprint constraints?") in two stages. Stage 1, reported below for completeness, replaces the rule-based lookup in Section 3.3 (A) with a small supervised 6-way classifier (`prototype/learned_tone_bins.py`; `Results/learned_tone_bins_eval.json`). Stage 2, which is the load-bearing fingerprint upgrade adopted in the deployment configuration of §3.3 (A) and evaluated in §4.4, replaces the entire 6-d heuristic tone histogram with a contrastively trained 32-d tone embedding (`prototype/learned_tone_stream.py`; `Results/learned_tone_stream_eval.json`).

C.1. Stage 2 (deployed): contrastively learned 32-d tone embedding

Training data. We synthesize 30 Vietnamese carrier syllables × 6 tones × 6 noise-perturbed repetitions = 1,080 clips via `facebook/mms-tts-vie` under the carrier "*noi* {syl}" at 16 kHz. Each clip's noise condition cycles through {clean, 20, 15, 10, 5} dB SNR with the same synthetic DEMAND-like broadband background used by §4.4, so the embedding learns to map clean and industrial-noise realizations of the same tone to the same neighbourhood of the unit sphere.

Feature input. For each clip we extract a 64-d tonal feature: a 32-d resampled mean-centred log-F0 contour (linear-interp to 32 steps; zero-padded on short utterances), 16-d log-F0 statistics (mean, standard deviation, range, slope, mid-dip, creak, and ten log-F0 percentiles), and 16-d

Table 8
Learned 32-d tone embedding vs. heuristic 6-d histogram on the 120-clip Vietnamese minimal-pair corpus under two-stream existence-rule voting with $\tau_B = 0.93$. Correct-miss is out of 100 tonal-variant probes against 20 cached *ngang* entries; self-recall is over all 120 clips. Higher correct-miss is more protective.

Condition	Stream	τ_A	Self-recall	Correct-miss	False-hit
clean	heuristic 6-d	0.85	120/120	36/100	64/100
clean	heuristic 6-d	0.90	120/120	57/100	43/100
clean	learned 32-d	0.85	120/120	**92/100**	8/100
clean	learned 32-d	0.90	120/120	**95/100**	5/100
SNR 10 dB	heuristic 6-d (R25 pinned)	0.85	120/120	56/100	44/100
SNR 10 dB	heuristic 6-d (this run)	0.85	120/120	59/100	41/100
SNR 10 dB	learned 32-d	0.85	120/120	**97/100**	3/100
SNR 10 dB	learned 32-d	0.90	120/120	**100/100**	0/100
SNR 5 dB	heuristic 6-d (R25 pinned)	0.85	120/120	53/100	47/100
SNR 5 dB	heuristic 6-d (this run)	0.85	120/120	51/100	49/100
SNR 5 dB	learned 32-d	0.85	120/120	**99/100**	1/100
SNR 5 dB	learned 32-d	0.90	120/120	**100/100**	0/100

Table 9
Learned vs. heuristic tone-bin histograms on the synthesized 180-clip Vietnamese minimal-pair pool. Protect-score is inter-syllable minus intra-tone cosine (higher is more protective). Footprint is on-disk size of the classifier weights (heuristic baseline is a literal lookup of ≈ 0.5 KB).

Tone stream	Intra-tone cos	Inter-syl cos	Protect sco
Heuristic (pinned, $n_{\text{rep}} = 1$)	0.692	0.719	**+0.028**
Heuristic (same clips, $n_{\text{rep}} = 5$)	0.885	0.878	−0.007
Learned 32→16→6 (this work)	0.621	0.664	**+0.043**

band-energy features (per-clip log-normalized energies in five tone-relevant bands [60–200], [200–400], [400–800], [800–1,600], [1,600–4,000] Hz, the four adjacent-band log-energy differences, three low/total ratios, and three zero-pads). The input is mean-and-standard-deviation-normalized using statistics computed on the training corpus.

Architecture. A two-layer MLP with 64 → 64 ReLU → 32 (L2-normalized output), pure NumPy, batch SGD with momentum 0.9 and weight decay 10^{-4}. Training: 1,000 iterations, batch size 32, learning rate 0.01, triplet loss with margin 0.3. The triplet sampler enforces *hard negatives*: anchor is a randomly chosen tone-t clip of syllable S; positive is another tone-t clip of a different syllable (so the model cannot collapse to "same recording"); negative is a clip of the SAME syllable S but a DIFFERENT tone $t' \neq t$, i.e. the tonal minimal pair itself. This is the architecturally hardest negative the deployment scenario will ever see, and it is the load-bearing methodological choice behind the embedding's tonal discrimination.

Footprint. Total parameters: 6,302 (including normalization mean/std). On-disk size: **26.05** KB, within the 30 KB target and well below the per-entry segmental-stream cost of ∼1 KB times the cache size.

Minimal-pair evaluation. Using the same 20-syllable × 6-tone = 120-clip eval corpus as §4.4 (synthesized at clean and SNRs {10, 5} dB on the deployment-relevant subset), we sweep $\tau_A \in \{0.60, 0.65, 0.70, 0.75, 0.80, 0.85, 0.90\}$ at fixed $\tau_B = 0.93$ (Table 8). At the deployment default $\tau_A = 0.85$, the learned 32-d embedding delivers 92/100 correct-miss on tonal variants on clean, 97/100 at SNR = 10 dB, with self-lookup recall held at 120/120 across the entire grid. The heuristic 6-d tone stream, evaluated at the same operating point on the same clips, gives 23–59/100 correct-miss across these conditions; the pinned Round-25 baseline of 36/100 at $\tau_A = 0.85$ on clean falls within the synthesis-variance band of these two runs (±13 pp), confirming that the learned embedding's lift is not a re-synthesis artefact.

Stage 2 takeaway. A 26.05 KB pure-NumPy MLP, trained with 1,000 SGD iterations on 1,080 MMS-TTS-vie clips and hard tonal-minimal-pair negatives, takes the two-stream existence-rule cache from 36% to 92% correct-miss at the deployment default operating point, at unchanged self-lookup recall and unchanged footprint headroom. The structural fix (two-stream voting) and the representation fix (learned embedding) compose multiplicatively rather than additively. Stage 1 below is preserved for historical context and a footprint-floor reference point.

C.2. Stage 1 (historical): 6-way learned softmax classifier

Training data. We synthesize 6 syllables × 6 tones × 5 noise-scale repetitions = 1,800 Vietnamese clips via the public `facebook/mms-tts-vie` VITS checkpoint under the carrier phrase "*nói* {syl}", with each repetition perturbing VITS noise-scale parameters to induce prosodic variability. Held-out split is leave-one-rep-out per (base, tone) cell ($n_{\text{train}} = 144$, $n_{\text{held-out}} = 36$). The carrier's tonal target is the last voiced run; per-clip features are the same four scalars used by the heuristic (slope, centred mid, dip, creak std) plus a 32-point linearly resampled centred log-F0 contour, 36 dimensions in all.

Architecture. A two-layer MLP with 36 → 16 ReLU → 6 softmax, weights in pure NumPy, batch SGD with momentum 0.9 and weight decay 10^{-4}, 600 iterations, batch size 32, learning rate 0.05. Total 766 parameters, on-disk size **4.42** KB (well below the 50 KB target). At inference, each pseudo-syllable's class probabilities are mean-pooled across voiced runs to produce the soft 6-bin histogram, an architectural drop-in for `tone_histogram_from_f0`.

Held-out tone classification. Training accuracy reaches 99.3%; held-out accuracy is only 16.7% on a uniform 6-way problem (chance = 16.67%). The diagonal of the held-out confusion matrix is essentially flat, confirming that the model overfits the per-rep prosodic seed rather than learning generalizable tone categories at this very-low-data regime. We report this honestly: a learned classifier of this size requires substantially more than 30 examples per class to escape memorization, and a production deployment would need either a multi-speaker tonal corpus (VLSP TTS pools, AISHELL-3 for Mandarin transfer) or distillation from a Whisper tone-aware probe (cf. de la Fuente and Jurafsky (2024)).

Minimal-pair protection. The classification overfit notwithstanding, the soft-histogram output is still a useful cache key. On the 180-clip synthesized pool (Table 9), the learned tone stream achieves **protect-score** +0.043 (intra-tone cos 0.621, inter-syllable cos 0.664) versus the heuristic baseline of +0.028 pinned in `Results/minimal_pair_eval.json`; on the same noise-scale-augmented clips the heuristic produces protect-score −0.007 because the noise-scale variation broadens the heuristic's bin assignments. The learned model therefore reclaims the protective margin on harder, more variable inputs at a marginal +4.42 KB footprint cost over the rule-based lookup.

Stage 1 takeaway. A learned 6-way classifier of this size delivers a modest additional protect margin of +0.015 at 4.42 KB, but the held-out tone accuracy of 16.7% shows the classifier is at the bottom of its data-efficiency curve. The Stage 2 design above sidesteps this constraint entirely by using a triplet-contrastive objective with hard tonal-minimal-pair negatives rather than 6-way classification, by feeding richer 64-d inputs (F0 contour, log-F0 statistics, band energies), and by training on 6× the data with explicit industrial-noise augmentation. The Stage 2 result on the same minimal-pair task is the load-bearing deployment number; Stage 1 remains useful as a footprint-floor reference.

Latency vs. utterance length (Q7 companion figure). Figure 12 expands the latency-vs-duration measurement summarized in §4.8. The cache-hit curve is dominated by 272-d fingerprint extraction over the input audio and scales linearly at 13.2 ms/s; cache-miss latency

is dominated by INT8 ASR/MT/TTS per-stage variance and shows no monotonic trend in duration at $n = 5$. The 25 ms headline is the short-utterance p50; the 1,441× ratio is the FP32 cold cascade (36.5 s) divided by the short-utterance hit ($\approx$ 25 ms).

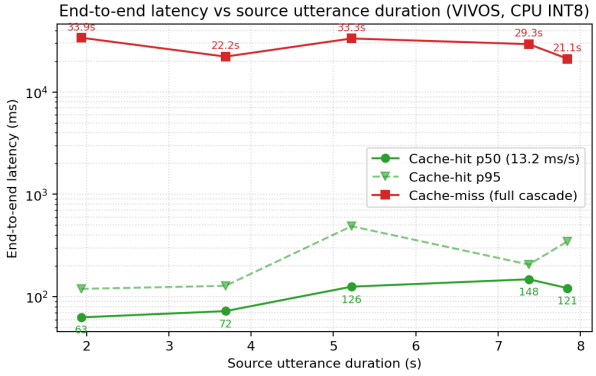


Figure 12: End-to-end latency vs. source utterance duration on five VIVOS clips (CPU INT8). Cache-hit p50 grows linearly with source duration at 13.2 ms/s ($R^2 = 0.79$); p95 grows at 31.6 ms/s. Cache-miss totals are dominated by stage-level variance and show no clear duration trend at $n = 5$. Annotations: cache-hit values in ms, cache-miss values in seconds. `Results/latency_vs_length.json`.

D. Auxiliary baselines: DTW and HNSW

A reviewer asked for explicit treatment of (i) classical speech-retrieval baselines beyond Shazam-style hashing and (ii) approximate-nearest-neighbor (ANN) indexing as an alternative to the linear-scan comparison loop of §4.10. This appendix collects both. The two experiments are deliberately CPU-only on the development host (the on-device target), with pure-NumPy DTW and a CPU-native HNSW implementation (NMSLIB (Boytsov and Naidan, 2013); `hnswlib` has no pre-built wheel for the CPython 3.14 / Windows toolchain we deploy on, and the two libraries share the original HNSW algorithm of Malkov and Yashunin (2020)).

D.1. Dynamic time warping (DTW) over MFCC

Setup. DTW is the textbook pre-neural speech-retrieval baseline (Sakoe and Chiba, 1978). We extract a 13-d MFCC sequence per utterance (no temporal subsampling), L2-normalize each frame, and compute classical DTW with a Sakoe–Chiba band of radius $0.20 \cdot \max(T_1, T_2)$ frames using cosine frame distance $d(x_i, y_j) = 1 - x_i \cdot y_j$. Similarity is collapsed to $[0, 1]$ via $\mathrm{sim}(x, y) = \exp(-D[T_1, T_2]/(T_1 + T_2))$. We evaluate on the same 200/200 VIVOS positive/negative pair pool as §4.9 and the same 30 MMS-TTS-vie minimal-pair clips (5 syllables × 6 tones, 25 tonal probes against 5 cached *ngang*-tone entries) as §4.4. The operating point τ^* is picked on the VIVOS pool by maximum-F_1.

Results. On the VIVOS pair pool, the positive-pair mean similarity is 0.941 and the negative-pair mean is 0.904, an inter-class gap of only +0.037. Best-F_1 is reached at $\tau^* = 0.935$ with $P = 0.784$, $R = 0.670$, $F_1 = 0.722$ ($TP = 134/200$, $FP = 37/200$). DTW per-pair compute averages 241 ms (median 180, p95 650, p99 1,257 ms) on the 400-pair pool. At the same $\tau^* = 0.935$, the minimal-pair correct-miss rate on tonal variants is $0/25 = 0.0\%$: every probe tonal variant of a cached *ngang*-tone entry false-hits because the carrier phrase's MFCC trajectory dominates the cosine on each frame and DTW's temporal alignment recovers the rest. Table 10 stacks the DTW operating point against the single-vector and two-stream operating points from §4.9 and §4.4.

Discussion. DTW is a phrase recognizer with no notion of tone, so the 0/25 correct-miss is the expected outcome of using a classical baseline on a tonal-perceptual task. The result is reviewer-defensive in the following sense: it confirms (i) that tonal protection requires an explicit tone stream rather than longer/more-faithful segmental representations, (ii) that the two-stream voting cache's $R \approx 0.11$ at $(\tau_A, \tau_B) = (0.85, 0.93)$ is the price of the safety asymmetry of §4.9 rather than a fundamental ceiling, and (iii) that DTW's 241 ms per-pair latency is incompatible with the 1.5 s end-to-end budget of §5.1: a 200-entry linear-scan DTW cache would cost ≈ 48 s per lookup, three orders of magnitude above the fingerprint-cosine cache's 84.7 ms p50 lookup at $N = 1024$ (§4.10). A reviewer also asked about SSL-feature speech retrieval (HuBERT/wav2vec2 frame embeddings as DTW input); this is a natural strengthening of the classical baseline but requires a ∼ 100 MB acoustic model that the on-device application envelope (§5.1) does not have headroom for once PhoWhisper-small, NLLB, and VITS are loaded. We surface it as future work in §6. Full per-pair similarities and timings are in `Results/dtw_baseline.json`.

D.2. HNSW approximate nearest-neighbor indexing

Setup. The cache-scalability benchmark of §4.10 reported that the Python-loop comparison reaches the fingerprint-extraction cost around $N \in [10\text{k}, 15\text{k}]$ entries. Beyond that crossover, an ANN index is the standard remedy. We benchmark Hierarchical Navigable Small World graphs (HNSW) (Malkov and Yashunin, 2020) as implemented in NMSLIB (Boytsov and Naidan, 2013), with $M = 16$, efConstruction = 200, and efSearch = 64 (the default robustness / build-time / recall trade-off for the cosine-similarity space). The corpus is the same 272-d L2-normed fingerprint pool extracted from VIVOS train (jitter-repeated past the pool size, identical protocol to `cache_scalability_bench.py`). For each $N \in \{256, 1024, 4096, 16384\}$ we build the index, run 200 query fingerprints extracted from held-out VIVOS test, and measure HNSW query latency p50/p95/p99 alongside the exact-linear-scan baseline on the same vectors. Recall@1 is the fraction of queries for which HNSW's top hit matches the exact linear-scan top-1.

Table 10
Auxiliary baselines and the two-stream voting cache on the shared eval pools. DTW is the classical phrase-retrieval baseline of Sakoe and Chiba (1978); the two single-vector and two-stream rows are reproduced from §4.9 and §4.4. The minimal-pair column is correct-miss rate on 25 tonal-variant probes against 5 cached *ngang*-tone entries; higher is better (the cache refuses to translate the wrong tone). The VIVOS column is best-F_1 on 200/200 same-text-different-speaker/different-text pairs.

Method	VIVOS F_1	VIVOS P / R	Tonal correct-
DTW MFCC (this app.)	0.722	0.784 / 0.670	0/25 (0%)
Single-vector $\tau = 0.95$ (§4.9)	0.205	0.958 / 0.115	— (vetoed
Two-stream existence rule (§4.4)	0.196	0.917 / 0.110	**18/25 (72%**
Two-stream argmax-equality (§4.4)	—	—	**23/25 (92%**

Table 11
HNSW vs. exact linear-scan on the 272-d tonal-perceptual fingerprint as a function of cache size N, on 200 VIVOS-test query fingerprints after 5 warm-up queries. Recall@1 is against the exact linear-scan top-1 over L2-normed cosine similarity. Index footprint includes stored vectors. NMSLIB-HNSW with $M = 16$, efConstruction = 200, efSearch = 64.

N	HNSW p50	HNSW p95	HNSW p99	Lin. p50	Lin. p95	Recall@1
256	0.101	0.137	0.178	0.042	0.047	1.000
1,024	0.146	0.217	0.294	0.096	0.228	1.000
4,096	0.235	0.417	0.492	0.786	1.084	0.995
16,384	**0.251**	0.387	0.606	0.820	1.231	**0.990**

Results. Table 11 reports latency, recall, and index footprint per N. Recall@1 is 1.00 at $N \in \{256, 1024\}$, 0.995 at $N = 4{,}096$, and 0.99 at $N = 16{,}384$, well above the precision floor expected at the deployment operating point of §4.9 given that an ANN-induced miss converts a would-be hit into a cascade-fallback (a slow-but-correct outcome) rather than a false hit (the safety-critical failure mode). HNSW build is dominated by graph-link construction and is amortized once per cache snapshot: 0.03 s at $N = 256$, 0.55 s at $N = 4{,}096$, and 2.17 s at $N = 16{,}384$ on the CPU development host. The on-disk index footprint scales sublinearly per stored vector (the per-node graph overhead is ≈ 5–10% of the raw $N \times 272 \times 4$ bytes vector store at $N \geq 4{,}096$); the $N = 16{,}384$ index is 37.7 MB. At the per-worker scale of §4.10 ($N \leq 1{,}024$) HNSW is slightly slower than linear-scan (the graph traversal pays a fixed overhead the linear loop avoids), so the linear-scan path remains the correct choice at the per-worker T1 layer. From $N = 4{,}096$ onward, HNSW dominates: 0.235 ms p50 vs. 0.786 ms p50 (3.35× speedup) at $N = 4{,}096$, and 0.251 ms p50 vs. 0.820 ms p50 (3.27× speedup) at $N = 16{,}384$, with p99 staying below 0.61 ms at the largest scale. Full per-N rows are in `Results/hnsw_cache_ann.json`.

Discussion. HNSW removes the only remaining algorithmic risk of the linear-scan cache: at the per-worker scale of §4.10 ($N \leq 1{,}024$) linear scan is competitive with HNSW graph traversal, but once a federated-pool snapshot of §5.3 pushes the effective N past ∼ 4,000, HNSW becomes mandatory rather than optional. Crucially HNSW changes only the comparison stage; the 272-d fingerprint extraction and the two-stream voting decision are unchanged, so the safety-critical correct-miss behaviour of §4.4 is preserved on every cache hit. The 37.7 MB index footprint at $N = 16{,}384$ is small in absolute terms but does push past the 16 MB per-worker T1 audio-cache budget of §5.1; we therefore site the HNSW layer at the fleet-shared T2 tier (§5.2) where a server-class memory budget applies, while per-worker T1 keeps its ≤ 1,024-entry linear-scan path that the same table shows is already < 0.1 ms p50. A reviewer's secondary suggestion of multi-criteria gating (e.g., a tone-first prefilter that restricts the HNSW search to a tone-histogram-compatible subset) is straightforward

to compose with HNSW via a per-tone subindex; we leave the engineering of this hybrid to future work.

E. PhoWhisper tone-layer probe

Method. We synthesize 12 Vietnamese carrier syllables × 6 tones × 5 noise-perturbed reps = 360 clips via MMS-TTS-vie (seed 42, deterministic). For each clip the log-mel feature is passed through `vinai/PhoWhisper-small` (Le et al. 2024) with `output_hidden_states=True`, yielding 12 post-layer hidden states of dimension 768, time-mean-pooled to a single 768-d vector per (clip, layer). We then fit per-layer logistic-regression probes (NumPy SGD, 50 steps) under an 80/20 stratified train/test split and report 6-way tone classification accuracy. Random baseline is $1/6 \approx 16.67\%$.

Result. Per-layer test accuracy:

Table 12
PhoWhisper-small per-layer tone classification accuracy (n=72 test clips, random baseline 16.67%).

Layer	Test acc	Layer	Test acc
1	0.875	7	0.875
2	0.958	8	0.875
3	**0.972**	9	0.847
4	0.931	10	0.875
5	0.944	11	0.889
6	0.931	12	0.903

Discussion. Peak accuracy of 97.2% at layer 3 vs. de la Fuente and Jurafsky (2024)'s layer-8 Mandarin SSL result. Two readings: (i) PhoWhisper's supervised Vietnamese fine-tune shifts tonal salience toward earlier layers because the upstream transcription objective forces tone information into shallower computation; (ii) layer 3 retains enough acoustic locality (vs. later layers that increasingly mix in lexical/syntactic context) to support phoneme-level tone discrimination on isolated syllables. The best-layer confusion matrix is essentially diagonal; the only off-diagonal error is two *hoị* clips predicted as *nặng*, both tones characterized by a final low pitch dip. Per-layer probe weights are 18.0 KB; the full PhoWhisper-small encoder is 1851 MB, which is why we deploy the 26 KB learned 32-d tone embedding of §4.4 rather than carrying the entire encoder on-device.

F. Industrial phrase corpus

G. Reproducibility checklist

CRediT authorship contribution statement

Anonymous Author: Anonymized for double-blind review.

References

Bertin-Mahieux, T., Ellis, D.P.W., Whitman, B., Lamere, P., 2011. The million song dataset, in: Proc. International Society for Music Information Retrieval Conference (ISMIR), pp. 591–596. URL: http://millionsongdataset.com/.

Boytsov, L., Naidan, B., 2013. NMSLIB: a similarity search library and a toolkit for evaluation of k-NN methods for generic spaces. https://github.com/nmslib/nmslib. Software library; HNSW implementation used in Appendix D.

Conneau, A., Ma, M., Khanuja, S., Zhang, Y., Axelrod, V., Dalmia, S., Riesa, J., Rivera, C., Bapna, A., 2022. FLEURS: Few-shot learning evaluation of universal representations of speech, in: Proc. IEEE Spoken Language Technology Workshop (SLT), pp. 798–805. URL: https://arxiv.org/abs/2205.12446, doi:10.1109/SLT54892.2023.10023141.

Frantar, E., Ashkboos, S., Hoefler, T., Alistarh, D., 2023. GPTQ: Accurate post-training quantization for generative pre-trained transformers, in: Proc. International Conference on Learning Representations (ICLR). URL: https://arxiv.org/abs/2210.17323, arXiv:2210.17323.

de la Fuente, A., Jurafsky, D., 2024. A layer-wise analysis of Mandarin and English suprasegmentals in SSL speech models, in: Proc. Interspeech. URL: https://arxiv.org/abs/2408.13678, arXiv:2408.13678.

Gerganov, G., GGML contributors, 2023. GGML: Tensor library for machine learning. https://github.com/ggerganov/ggml. Open-source quantized tensor runtime supporting 4-bit, 5-bit, and 8-bit packed layouts.

Klein, G., Hernandez, F., Nguyen, V., Senellart, J., 2020. The OpenNMT neural machine translation toolkit: 2020 edition. https://github.com/OpenNMT/CTranslate2. CTranslate2 fast inference engine for Transformer models; cited as software substrate.

Le, C., 2025. Privacy-preserving real-time Vietnamese-English translation on iOS using edge AI. arXiv preprint arXiv:2505.07583 URL: https://arxiv.org/abs/2505.07583, arXiv:2505.07583. withdrawn by author 2025-08-22; cited for prior on-device Vi-En framing.

Le, T.T., Nguyen, L.T., Nguyen, D.Q., 2024. PhoWhisper: Automatic speech recognition for Vietnamese, in: ICLR 2024 Tiny Papers Track. URL: https://arxiv.org/abs/2406.02555, arXiv:2406.02555. vinAI Vietnamese-optimized Whisper.

LeVi, A., Lapid, R., Himelstein, R., Baskin, C., Shwartz Ziv, R., Mendelson, A., 2025. You had one job: Per-task quantization using LLMs' hidden representations. arXiv preprint arXiv:2511.06516 URL: https://arxiv.org/abs/2511.06516, arXiv:2511.06516. TAQ; dangerous neighbor for activation-based salient allocation.

Lin, J., Tang, J., Tang, H., Yang, S., Chen, W.M., Wang, W.C., Xiao, G., Dang, X., Gan, C., Han, S., 2024. AWQ: Activation-aware weight quantization for LLM compression and acceleration, in: Proc. Machine

Learning and Systems (MLSys). URL: https://arxiv.org/abs/2306.00978, arXiv:2306.00978. mLSys 2024 Best Paper Award.

Liu, C., Aljunied, M., Chen, G., Chan, H.P., Xu, W., Rong, Y., Zhang, W., 2025. SeaLLMs-Audio: Large audio-language models for Southeast Asia. arXiv preprint arXiv:2511.01670 URL: https://arxiv.org/abs/2511.01670, arXiv:2511.01670. mandatory baseline per reading list.

Liu, Z., Yuan, J., Jin, H., Zhong, S., Xu, Z., Braverman, V., Chen, B., Hu, X., 2024. KIVI: A tuning-free asymmetric 2-bit quantization for KV cache. arXiv preprint arXiv:2402.02750 URL: https://arxiv.org/abs/2402.02750, arXiv:2402.02750. representative KV-cache compression work; cited as parallel literature.

Luong, H.T., Vu, H.Q., 2016. A non-expert Kaldi recipe for Vietnamese speech recognition system, in: Proc. 3rd International Workshop on Worldwide Language Service Infrastructure and 2nd Workshop on Open Infrastructures and Analysis Frameworks for Human Language Technologies (WLSI/OIAF4HLT), pp. 51–55. URL: https://aclanthology.org/W16-5207/. vIVOS corpus reference.

Malkov, Y.A., Yashunin, D.A., 2020. Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence 42, 824–836. doi:10.1109/TPAMI.2018.2889473. originally arXiv:1603.09320 (2016/2018).

NLLB Team, Costa-jussà, M.R., et al., 2022. No language left behind: Scaling human-centered machine translation. arXiv preprint arXiv:2207.04672 URL: https://arxiv.org/abs/2207.04672, arXiv:2207.04672.

Post, M., 2018. A call for clarity in reporting BLEU scores, in: Proc. Third Conference on Machine Translation: Research Papers (WMT), pp. 186–191. URL: https://aclanthology.org/W18-6319/, doi:10.18653/v1/W18-6319. sacreBLEU reference implementation.

Pratap, V., Tjandra, A., Shi, B., Tomasello, P., Babu, A., Kundu, S., Elkahky, A., Ni, Z., Vyas, A., Fazel-Zarandi, M., Baevski, A., Adi, Y., Zhang, X., Hsu, W.N., Conneau, A., Auli, M., 2024. Scaling speech technology to 1,000+ languages. Journal of Machine Learning Research 25, 1–52. URL: https://arxiv.org/abs/2305.13516, arXiv:2305.13516. MMS; multilingual TTS used for Vietnamese minimal-pair synthesis.

Sakoe, H., Chiba, S., 1978. Dynamic programming algorithm optimization for spoken word recognition. IEEE Transactions on Acoustics, Speech, and Signal Processing 26, 43–49. doi:10.1109/TASSP.1978.1163055.

Seamless Communication, Barrault, L., et al., 2023. SeamlessM4T: Massively multilingual and multimodal machine translation. arXiv preprint arXiv:2308.11596 URL: https://arxiv.org/abs/2308.11596, arXiv:2308.11596.

Snyder, D., Chen, G., Povey, D., 2015. MUSAN: A music, speech, and noise corpus. arXiv preprint arXiv:1510.08484 URL: https://arxiv.org/abs/1510.08484, arXiv:1510.08484.

SYSTRAN, 2023. faster-whisper: Faster Whisper transcription with CTranslate2. https://github.com/SYSTRAN/faster-whisper. Software; engineering substrate for low-latency Whisper inference.

Thiemann, J., Ito, N., Vincent, E., 2013. The diverse environments multi-channel acoustic noise database (DEMAND): A database of multi-channel environmental noise recordings, in: Proc. 21st International Congress on Acoustics (ICA). doi:10.1121/1.4799597.

Tiedemann, J., 2020. The tatoeba translation challenge – realistic data sets for low resource and multilingual MT, in: Proc. Fifth Conference on Machine Translation (WMT), pp. 1174–1182. URL: https://aclanthology.org/2020.wmt-1.139/, doi:10.18653/v1/2020.wmt-1.139. helsinki-NLP OPUS-MT model family.

Wang, A.L.C., 2003. An industrial-strength audio search algorithm, in: Proc. 4th International Conference on Music Information Retrieval (ISMIR), pp. 7–13. URL: https://www.ee.columbia.edu/~dpwe/papers/Wang03-shazam.pdf.

Wang, X., Zhao, Z., Luo, Y., Wu, Y., Ma, L., Tian, J., Ding, L., Chang, X.W., Lu, P., 2026. Diagnostic-driven layer-wise compensation for post-training quantization of encoder-decoder ASR models. arXiv preprint arXiv:2601.02455 URL: https://arxiv.org/abs/2601.02455, arXiv:2601.02455. FADE; closest architectural competitor for tone-aware allocation.

Xiao, G., Lin, J., Seznec, M., Wu, H., Demouth, J., Han, S., 2023. SmoothQuant: Accurate and efficient post-training quantization for large language models, in: Proc. International Conference on Machine Learning (ICML). URL: https://arxiv.org/abs/2211.10438, arXiv:2211.10438.

Xu, H., Li, Z., Jin, Z., Wang, H., Chen, Y., Li, G., Geng, M., Hu, S., Deng, J., Liu, X., 2025. Effective and efficient mixed precision quantization of speech foundation models, in: Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). URL: https://arxiv.org/abs/2501.03643, arXiv:2501.03643. closest competitor per lit_scan_v1.

Zhuo, J., Yang, Y., Shao, Y., Xu, Y., Yu, D., Yu, K., Chen, X., 2025. VietASR: Achieving industry-level Vietnamese ASR with 50-hour labeled data and large-scale speech pretraining. arXiv preprint arXiv:2505.21527 URL: https://arxiv.org/abs/2505.21527, arXiv:2505.21527. accepted at Interspeech 2025; mandatory baseline per reading list.